Business Master Checklist: legal, finance, sales, ops, and compliance roadmap for taking a tech product to market

Computer
Go-To-Market Playbook: sales funnel, pricing tiers, customer acquisition, and marketing infrastructure for a new company
Computer
Show me the exact legal and compliance checklist
Give me the go-to-market pilot sales playbook
Provide the customer success and onboarding workflow

To take Semester to market, build three systems in parallel: a **legal-and-trust system** that lets institutions buy safely, a go-to-market system that creates and closes measurable pilots, and a customer-success system that turns pilots into renewals and case studies. Do not treat the checklists below as legal or compliance advice—use them as an execution and evidence checklist for qualified counsel, accountants, insurers, and security reviewers.

For an education-focused SaaS company, the minimum credible launch posture is: registered company, tax/banking/accounting setup, ownership and contract pack, privacy/security controls, an institutional pilot offer, a sales pipeline, onboarding and support, and evidence that customers achieve the promised outcome. The SBA and IRS both identify entity structure, registration, EIN/tax obligations, bank account, licenses, and insurance among foundational launch steps. [1] [2]

Legal and compliance checklist

A. Company formation and ownership

Item	What to create or verify	Owner	Launch gate
Legal entity	LLC or corporation, state registration, registered agent, governing documents	Founder + counsel	Entity exists and is in good standing
EIN and tax accounts	Federal EIN, relevant state/local registrations, tax-year decision	Founder + accountant	Can invoice, pay taxes, hire, and open accounts
Business bank account	Dedicated business checking, payment controls, bookkeeping feed	Founder + finance	No company money runs through personal accounts
Corporate records	Formation documents, consents, ownership ledger, board approvals	Founder + counsel	Central, current, access-controlled record

Item	What to create or verify	Owner	Launch gate
Cap table	Founder equity, options, vesting, advisor grants, dilution model	Founder + counsel	Single source of truth
IP assignment	Founder, employee, contractor, advisor invention/IP assignment agreements	Founder + counsel	Company owns code, designs, content, data models, and trademarks
Contractor agreements	Scope, confidentiality, IP assignment, payment, security obligations	Founder + counsel	Every contractor is covered before work starts
Trademark strategy	Name search, classes, filing decision, domain/social ownership inventory	Founder + counsel	Brand ownership risk understood
Business insurance	Cyber, E&O/professional liability, general liability; evaluate D&O as company grows	Founder + broker	Coverage matches customer and contract requirements

The SBA notes that a registered business typically needs a business structure, registration, tax IDs, bank account, required licenses/permits, and insurance appropriate to the business and jurisdiction; state obligations vary.[1]

B. Product contracts

Create a controlled contract library. Every template needs: owner, version, last review date, legal-review status, usage instructions, and approved fallback language.

Document	Why it matters	Use it when
Mutual NDA	Protects two-way confidential discussions	Early institutional, partner, or investor conversations
One-way NDA	Protects Semester confidential materials	Demos, contractors, evaluators
Master Services Agreement	Core commercial/legal relationship	Institutional or enterprise customer contracts
SaaS Subscription Agreement	Product-specific service, subscription, payment, and use terms	Standard SaaS purchase
Order form	Price, term, scope, users, billing, renewal, special terms	Every paid customer agreement
Pilot agreement	Defines limited pilot, outcomes, conversion, and end date	Initial institutional deployment
Statement of Work	Defines implementation or professional services	Paid setup, migration, training, integration work
Data Processing Addendum	Defines data-processing roles and safeguards	Customer handles personal data
Service Level Agreement	Defines support, uptime, severity, response, and remedies	Enterprise/institutional procurement
Acceptable Use Policy	Defines prohibited conduct and platform boundaries	Public product terms and customer contracts
Privacy Policy	Explains data practices to users	Website/product launch

Document	Why it matters	Use it when
Terms of Service	Governs use for self-service or individual users	Public-facing product access
Cookie notice	Explains analytics/marketing cookies and choices	Website if applicable
Subprocessor list	Names material vendors that process customer data	Institutional procurement and trust center
Accessibility statement	Explains accessibility commitment, support path, and conformance posture	Customer-facing trust materials
Security addendum	Provides verified security commitments	Security review or enterprise contract

C. Privacy and data governance

For Semester, define this before importing student, institutional, or staff data.

- [ ] Data inventory: every field, source, owner, purpose, legal basis/contractual reason, storage location, retention period, and recipients.
- [ ] Data-flow diagrams: browser → API → database → analytics → support → backups → subprocessors.
- [ ] Data classification: public, internal, confidential, restricted/student-sensitive.
- [ ] Data minimization policy: collect only data necessary for the defined workflow.
- [ ] Retention schedule: define default retention, customer-configurable periods, archive behavior, and deletion verification.
- [ ] Data deletion process: account deletion, customer termination, backup expiration, support-ticket data, exports, and logs.
- [ ] User rights/request process: access, correction, deletion, and escalation workflow as applicable.
- [ ] Privacy-policy accuracy review: statements must match actual product behavior and vendors.
- [ ] Subprocessor approval process: security/privacy review before adding a vendor.
- [ ] Marketing-consent process: email/SMS preferences, unsubscribe, and suppression lists.
- [ ] Data-breach triage and notification process.
- [ ] Signed DPA workflow for institutional customers.
- [ ] Cross-border transfer review if customer or user data leaves the United States.
- [ ] AI-data policy if customer/student data is sent to any AI model provider.

The FTC's business guidance emphasizes a security plan that collects only necessary information, safeguards retained data, and disposes of data securely; these are operational practices as well as trust requirements.[3]

D. Security and compliance

Start with an evidence-based baseline. Do not claim “SOC 2 compliant,” “FERPA compliant,” “HIPAA compliant,” “WCAG compliant,” or similar unless the claim is factually supportable and appropriately reviewed.

Security baseline

- [ ] Asset inventory: systems, domains, repositories, cloud accounts, devices, services, and data stores.
- [ ] Security owner and escalation contacts.
- [ ] Written information-security program.
- [ ] Access-control policy with least privilege.
- [ ] MFA required for all production/admin systems.
- [ ] Formal joiner/mover/leaver access process.
- [ ] Centralized secrets management.
- [ ] No production secrets in source control, browser code, screenshots, or documentation.
- [ ] Encryption in transit and at rest—document actual provider/configuration evidence.
- [ ] Secure development lifecycle.
- [ ] Pull-request review and branch-protection policy.
- [ ] Dependency scanning and patch workflow.
- [ ] Vulnerability intake, severity, SLA, remediation, and retest workflow.
- [ ] Production logging and error monitoring.
- [ ] Backup and restore procedure, tested on a defined schedule.
- [ ] Incident-response plan with contacts, severity definitions, communication templates, and tabletop test.
- [ ] Business continuity/disaster-recovery plan.
- [ ] Vendor-risk assessment process.
- [ ] Security training and acknowledgement process.
- [ ] Security evidence register.
- [ ] Public Trust Center or controlled security-document request flow.

Higher-ed vendor readiness

- [ ] HECVAT Lite response.
- [ ] Full HECVAT preparation for higher-risk or enterprise deployments.
- [ ] Accessibility review and VPAT roadmap, if requested by institutions.
- [ ] Security questionnaire answer library.
- [ ] Architecture diagram.

- [ ] Data-flow diagram.
- [ ] Penetration-test strategy or qualified external-testing plan.
- [ ] Third-party vulnerability disclosure or security contact process.
- [ ] Customer-facing security overview.
- [ ] Verified subprocessor list.
- [ ] Data-retention/deletion documentation.
- [ ] Incident-notification commitments in contract language.

HECVAT is a higher-education vendor questionnaire framework for evaluating security and privacy posture. HECVAT Lite is generally suited to lower-risk reviews, while Full is more comprehensive for sensitive or critical workloads. [4] [5]

E. Compliance review queue

Track every item in a register:

Field	Example
Requirement	DPA review
Area	Privacy
Trigger	Institution uploads student records
Owner	Security/privacy lead
Evidence	Signed DPA, data-flow map, subprocessors
Reviewer	Counsel or privacy adviser
Status	Draft / under review / approved / expired
Renewal date	Annual or contract-specific
Risk	High
Customer impact	Blocks pilot signature
Notes	Customer-specific deviations

Legal/compliance master prompt

```
You are Semester's legal operations, privacy operations, security compliance, and ver

Read all existing legal, security, privacy, data, contracts, Supabase, infrastructure

Do not generate duplicate policies. First create an evidence inventory showing:
- Existing artifact
- Path
- Owner
- Last update
- Current status
- Whether it is a draft, approved, customer-facing, internal-only, stale, duplicate,
```

```
- Related product/data flow
- Required reviewer: counsel, accountant, insurer, security, accessibility, executive

Then create or update a single compliance operating system covering:

1. Entity, ownership, IP assignment, contractor agreements, tax/accounting setup, ins
2. Commercial contract library:
   - NDA
   - MSA
   - SaaS agreement
   - order form
   - pilot agreement
   - SOW
   - DPA
   - SLA
   - acceptable use
   - privacy policy
   - terms
   - subprocessor list
3. Privacy program:
   - data inventory
   - data classification
   - data flows
   - retention schedule
   - deletion procedure
   - data request process
   - vendor/subprocessor review
   - AI data-use policy
4. Security program:
   - access control
   - MFA
   - secrets
   - secure development
   - dependency and vulnerability management
   - logging
   - backups
   - incident response
   - business continuity
   - disaster recovery
   - vendor risk
5. Higher-education procurement readiness:
   - HECVAT Lite roadmap
   - Full HECVAT roadmap
   - security questionnaire response library
   - Trust Center inventory
   - accessibility evidence roadmap
   - customer security evidence pack
6. Compliance register:
   - owner
   - risk
   - evidence
   - reviewer
   - status
   - renewal date
   - customer or launch impact
```

```
Do not make legal, privacy, security, accessibility, or compliance claims that lack v

Create:
- docs/business/compliance/COMPLIANCE_MASTER_REGISTER.md
- docs/business/compliance/TRUST_CENTER_CONTENT.md
- docs/business/compliance/HECVAT_LITE_ROADMAP.md
- docs/business/compliance/SECURITY_QUESTIONNAIRE_LIBRARY.md
- docs/business/compliance/DATA_INVENTORY.md
- docs/business/compliance/DATA_RETENTION_AND_DELETION.md
- docs/business/compliance/CONTRACT_LIBRARY_INDEX.md
- docs/business/compliance/INCIDENT_RESPONSE_AND_CUSTOMER_COMMS.md
- docs/business/compliance/VENDOR_RISK_REGISTER.md

Return the PO items that block a paid institutional pilot, evidence gaps, and exact h
```

Go-to-market pilot sales playbook

Semester should launch through a **narrow, measurable pilot**—not a vague "replace the entire university stack" sale. The first sale should solve a costly or frustrating workflow for a specific institutional buyer, with proof and a defined annual conversion decision.

A. Define the launch wedge

Use this one-page template:

```
Product:
Semester

Initial target segment:
[Example: mid-sized private universities, community colleges, student-success departm

Economic buyer:
[Named role, not "the university"]

Champion:
[Role that has daily pain and can mobilize internal stakeholders]

End users:
[Students, advisors, staff, faculty, administrators]

Current problem:
[Fragmented process, manual work, delays, missed handoffs, low student activation, po

Semester workflow:
[One job that Semester completes measurably better]

Pilot scope:
[One campus / one department / one student cohort / one term / a defined integration

Success metrics:
[Activation, completed student action, time saved, workflow completion, satisfaction

Pilot duration:
[Example: 8-12 weeks or one academic term]
```

```
Pilot commercial terms:
[Paid pilot amount, implementation scope, included seats/users, data limits, support

Annual conversion:
[Annual price, buyer decision date, conversion criteria, expected expansion path]
```

B. Sales process

Stage	Objective	Required output	Exit criteria
1. Targeting	Select accounts with real fit	Named account plan	Buyer/champion hypothesis exists
2. Outreach	Start relevant conversation	Personalized message and reason now	Reply, referral, or meeting
3. Discovery	Confirm urgent pain and buying path	Discovery notes and qualification score	Clear problem, stakeholders, timeline
4. Demo	Show the exact target workflow	Role-specific demo and follow-up	Buyer sees a plausible outcome
5. Solution design	Scope a pilot with the customer	Pilot design, integration/data assumptions	Champion agrees to success measures
6. Security/procurement	Reduce institutional risk	Trust pack, HECVAT/Security answers, DPA path	Security/legal/procurement path known
7. Proposal	Create commercial decision	Pilot proposal, pricing, mutual action plan	Decision date and owner named
8. Contracting	Obtain signed agreement	Pilot agreement/order form	Signed + implementation scheduled
9. Pilot delivery	Prove measurable value	Weekly success report	Metrics on track
10. Conversion	Turn proof into annual contract	Results deck and annual proposal	Renew/expand/close-lost decision

C. Account qualification

For each account, answer:

- Is the problem expensive, risky, or politically visible?
- Is there an identifiable champion?
- Is there an economic buyer?
- Can the buyer fund a pilot?
- Is there a current trigger:
  - retention pressure
  - new initiative
  - system dissatisfaction

    - consolidation project
    - accreditation/accessibility pressure
    - new leadership
    - student-service bottleneck
    - upcoming term/deadline
- Can the deployment be scoped without a multi-year integration project?
- Can Semester measure success within one term or one quarter?
- Can the customer become a reference/case study?
- Does the account fit your desired annual contract profile?

D. Discovery call script

```
Opening:
“Thank you for making time. I want to understand the current workflow,
where students or staff get stuck, what it costs, and whether a limited,
measurable Semester pilot is worth designing.”

Problem:
1. Walk me through how this works today.
2. Where do students, staff, or departments lose time?
3. What breaks most often?
4. What existing tools are involved?
5. What happens when the workflow fails?
6. How is success measured today?
7. What are the consequences of doing nothing?

Stakeholders:
8. Who feels this pain daily?
9. Who owns the budget?
10. Who will assess security, accessibility, privacy, procurement, and legal?
11. Who must approve a pilot?
12. Who could block it?

Commercial and timeline:
13. Is there a budget source or initiative for this?
14. What decision deadline matters?
15. What would make a pilot too risky or too difficult?
16. What proof would justify annual adoption?

Close:
“If Semester could improve [agreed outcome] for [defined population] by
[metric] within [pilot timeframe], would you be open to a scoped pilot
proposal with explicit success criteria and a conversion decision?”
```

E. Pilot proposal contents

Every pilot proposal should include:

- Executive summary
- Current-state problem
- Proposed Semester workflow
- Target users and departments
- Scope and non-goals
- Timeline
- Data/integration assumptions
- Implementation responsibilities
- Semester responsibilities
- Institution responsibilities
- Training/support plan
- Security/privacy/accessibility scope
- Success metrics and baseline
- Weekly operating cadence
- Risks and dependencies
- Pilot price
- Annual conversion price
- Conversion decision date
- Named decision maker
- Mutual Action Plan

F. Mutual Action Plan

Workstream	Customer owner	Semester owner	Deliverable	Due date	Status
Executive sponsorship	VP/Dean/CIO	Founder	Named sponsor and goals	Date	Not started
Security review	Security lead	Security owner	Questionnaire and evidence	Date	In progress
Data review	Privacy/IT	Engineering	Data map and DPA	Date	Not started
Contracting	Procurement/legal	Founder/counsel	Pilot agreement	Date	Not started
Implementation	Admin lead	CS/implementation	Configuration and training	Date	Not started

Workstream	Customer owner	Semester owner	Deliverable	Due date	Status
Launch	Champion	Customer success	User rollout	Date	Not started
Midpoint review	Executive sponsor	Founder	KPI check	Date	Not started
Conversion	Economic buyer	Founder	Annual decision	Date	Not started

G. Sales assets to produce

- [ ] Institutional one-pager
- [ ] Buyer-specific one-pagers
- [ ] Sales deck
- [ ] Demo script
- [ ] Demo environment
- [ ] Discovery guide
- [ ] Qualification scorecard
- [ ] ROI calculator
- [ ] Pilot proposal template
- [ ] Mutual Action Plan
- [ ] Pricing calculator
- [ ] Quote/order form template
- [ ] Security overview
- [ ] HECVAT Lite or readiness packet
- [ ] DPA/Privacy response
- [ ] Accessibility response
- [ ] Procurement FAQ
- [ ] Competitive battlecards
- [ ] Objection-handling library
- [ ] Case-study template
- [ ] Reference call guide
- [ ] Renewal/expansion deck

H. Pipeline metrics

Track weekly:

Metric	Why it matters
Named target accounts	Top-of-funnel capacity
New buyer conversations	Actual market learning
Qualified opportunities	Real pipeline, not vanity leads
Champion identified	Deal viability
Economic buyer identified	Commercial viability
Demo-to-pilot-design rate	Product/positioning fit
Security review initiated	Institutional seriousness
Pilot proposal count	Near-term pipeline
Pilot close rate	Offer and buying-motion quality
Time from discovery to signature	Sales cycle health
Pilot activation	Delivery quality
Pilot KPI attainment	Conversion readiness
Pilot-to-annual conversion	Product-market and commercial proof
Expansion pipeline	Long-term account value

Customer success and onboarding workflow

Customer success starts during sales—not after signature. The customer must know what success means before the pilot begins.

A. Handoff from sales to implementation

Trigger: Contract or pilot agreement signed.

Sales must provide:

- Account plan
- Buyer map
- Champion map
- Contract scope
- Pilot objectives
- Expected success metrics
- Known risks
- Procurement/security commitments

- Promised integrations
- Commercial deadlines
- Conversion date
- Stakeholder contact list
- Decision-making process
- All verbal commitments recorded in CRM

Customer success must confirm:

- Scope is implementable
- Required customer inputs are known
- Admins are identified
- Data/security needs are understood
- Timeline is realistic
- Success baseline can be measured
- Support/escalation contacts are set
- Training plan exists

B. Implementation workflow

Phase	Target timing	Customer outcome	Semester deliverable
Contract-to-kickoff	0–5 business days	Customer knows next steps	Welcome email, kickoff scheduling, implementation checklist
Kickoff	Week 1	Shared goals and owners	Kickoff deck, RACI, success plan, implementation plan
Configuration	Weeks 1–2	Workspace matches agreed scope	Tenant/configuration setup, access roles, workflows
Data/integrations	Weeks 1–3	Data is validated and safe	Data map, import/integration checklist, validation report
Admin training	Weeks 2–3	Administrators can operate system	Admin guide, recorded training, office hours
User rollout	Weeks 3–4	Users are invited and activated	Launch communications, onboarding, QR assets, support coverage
Adoption	Weeks 4–8	Core workflows are used	Adoption dashboard, nudges, help center, champion check-ins
Midpoint review	Mid-pilot	Risks resolved early	KPI report, action plan, executive update
Results review	Final 2 weeks	Value is documented	Outcomes report, ROI summary, annual proposal
Conversion/renewal	Before end date	Customer makes an informed decision	Renewal plan, expansion roadmap, reference/case-study request

C. Pilot success plan

```
Customer:
[Institution / department]

Pilot period:
[Start date] to [End date]

Executive sponsor:
[Name / role]

Operational champion:
[Name / role]

Semester owner:
[Name / role]

Target user population:
[Who is included]

Problem baseline:
[What happens today, including measurable pain]

Desired outcome:
[Outcome statement]

Primary metrics:
1. Activation rate
2. First meaningful action
3. Workflow completion
4. Weekly active use
5. Time saved or capacity gained
6. Error reduction or service-quality improvement
7. User/admin satisfaction

Baseline measurement:
[Current numbers and evidence source]

Target measurement:
[Success thresholds]

Cadence:
- Weekly operating meeting
- Biweekly adoption report
- Midpoint executive review
- Final results and conversion review

Conversion decision:
- Decision owner
- Decision date
- Annual scope
- Commercial proposal
- Required proof to approve annual agreement
```

D. Customer health score

Create a 0–100 health score using:

Signal	Example weight	What it detects
Admin engagement	15%	Operational ownership
User activation	20%	Whether rollout worked
Core workflow completion	20%	Actual product value
Weekly active usage	15%	Ongoing adoption
Support sentiment	10%	Friction or dissatisfaction
Executive engagement	10%	Renewal/conversion risk
Success-metric trend	10%	Real business outcome

Flag at-risk accounts when:

- Admin engagement drops
- Activation misses target
- Workflow completion declines
- Support tickets rise
- Executive sponsor stops attending
- Security/integration commitments stall
- No conversion meeting is scheduled

E. Support system

Create:

- Support intake form
- Support email or portal
- Severity levels:
  - P0: security/data loss/system outage
  - P1: critical workflow blocked
  - P2: degraded feature/workaround exists
  - P3: how-to/feature request
- Response and update targets
- Internal escalation path
- Incident communications template
- Known-issues page
- Knowledge base

- Admin help center
- Student-facing help content
- Product-feedback tag taxonomy
- Root-cause/postmortem template
- Support-to-product escalation workflow

F. Renewal and expansion

Start the renewal motion during the pilot—not in the final week.

Timing	Action
Contract signature	Record renewal date, conversion date, annual scope hypothesis
Week 1	Confirm success criteria and executive sponsor
Week 3	Check activation and implementation risks
Midpoint	Present preliminary outcomes and unblock issues
60–90 days before renewal	Confirm annual budget/process/decision owner
30–60 days before renewal	Present value report and annual proposal
Final month	Complete contracting, procurement, and expansion scope
After renewal	Launch expansion plan and request reference/case study

Customer-success execution prompt

```
You are the Head of Customer Success and Implementation for Semester.

Read the existing Semester product, security, support, implementation, pilot, analyti

Build an implementation-ready customer success system for institutional and departmer

Create:
1. Sales-to-CS handoff checklist
2. Customer kickoff process
3. Implementation project plan
4. Customer RACI
5. Data/integration readiness checklist
6. Admin configuration checklist
7. Training curriculum for administrators, staff, champions, and students
8. Student activation and campus launch kit
9. Support policy and ticket severity model
10. Escalation process
11. Customer health-score model
12. Pilot success plan template
13. Weekly pilot operating review
14. Midpoint executive review template
15. Final results/value report
16. QBR template
17. Renewal and expansion playbook
```

```
18. Churn-risk playbook
19. Customer feedback and product-request taxonomy
20. Case-study and reference-request process

For every artifact, define:
- Objective
- Owner
- Trigger
- Inputs
- Steps
- Output
- Customer communication
- Internal communication
- Success metrics
- Escalation condition
- Storage/system of record
- Review cadence

Prioritize one narrow, measurable institutional pilot that can produce a case study a

Return:
- P0 pilot-readiness gaps
- 30/60/90-day customer-success roadmap
- Required templates
- Customer health dashboard specification
- Renewal/expansion workflow
- Risks that could prevent pilot conversion
```

Your immediate build sequence

1. **Legal/tax/company setup:** entity, EIN, banking, accounting, founder/contractor IP assignments, insurance review.
2. **Trust and procurement:** security overview, data map, privacy policy, DPA, incident process, HECVAT Lite evidence plan.
3. **Narrow pilot offer:** one ICP, one workflow, one buyer, one success metric set, one commercial package.
4. **Sales system:** CRM, 100-account list, discovery/demo/proposal/pilot templates.
5. **Customer success system:** onboarding, implementation, support, pilot success, renewal process.
6. **Marketing infrastructure:** positioning, website, demo request, Trust Center, content and lifecycle email.
7. **Student adoption:** onboarding, activation definition, campus launch kit, ambassadors, retention dashboard.
8. **Operating cadence:** weekly pipeline/product/customer reviews; monthly finance/security reviews; quarterly planning.

The sequence matters: **trust → offer → pipeline → pilot delivery → proof → repeatable growth.**

⁂

1. https://www.sba.gov/counseling/launch-your-business/
2. https://www.irs.gov/businesses/small-businesses-self-employed/checklist-for-starting-a-business
3. https://www.ftc.gov/business-guidance/privacy-security/data-security
4. https://aws.amazon.com/blogs/publicsector/higher-education-community-vendor-assessment-toolkit-now-available-on-aws-artifact/
5. https://www.upguard.com/blog/hecvat
6. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit
7. https://cloud.google.com/security/compliance/hecvat
8. https://www.zoom.com/en/trust/legal-compliance/hecvat/
9. https://www.ren-isac.net/hecvat/index.html
10. https://www.purchasing.utah.edu/glossary/hecvat-higher-education-community-vendor-assessment-toolkit/
11. https://www.saltycloud.com/blog/what-is-the-hecvat/
12. https://www.ftc.gov/reports/federal-trade-commission-2023-privacy-data-security-update
13. https://www.ftc.gov/news-events/topics/protecting-consumer-privacy-security
14. https://www.linkedin.com/pulse/higher-education-community-vendor-assessment-tool-why-michael-sc3oe
15. https://www.ftc.gov/business-guidance/resources/protecting-personal-information-guide-business

Draft the master GTM playbook prompt

Generate the interactive financial model
Write the pilot sales email sequence
Build the compliance data security plan
Export this playbook as a PDF report

The GTM playbook PDF report has already been generated with the master execution prompt, financial-model specification, pilot sales structure, customer-success workflow, security/compliance plan, and launch checklist. I will not regenerate a duplicate file; below are the missing **pilot sales email sequence** and an implementation-ready **interactive model prompt** to use alongside the report.

The operating strategy is to sell a narrow, measurable institutional pilot first, then convert measured student/staff outcomes into an annual contract and expansion motion. Higher-ed buyers commonly require vendor-risk evidence; HECVAT is designed to support campus evaluation of vendor security, privacy, accessibility, and related controls.[1] [2] [3]

Master GTM playbook prompt

Use this prompt in Claude Code, v0, Cursor, or another planning/coding agent with access to the Semester repository:

```
You are the Chief Revenue Officer, Head of Marketing, Customer Success leader,
finance lead, security-program coordinator, and Chief of Staff for Semester.

Semester is an education-technology SaaS company serving institutions,
departments, students, and individual users. Its launch strategy uses two
connected growth engines:

1. Institutional B2B sales:
   Sell one narrow, measurable workflow through paid pilots that convert into
   annual platform contracts.

2. Student adoption:
   Drive activation, first meaningful action, workflow completion, retention,
   satisfaction, and peer advocacy inside each institutional deployment.

Read all existing Semester product, engineering, security, privacy, financial,
launch, pricing, sales, customer-success, operations, and audit documentation
before generating new work.

Do not create duplicate artifacts. First produce an inventory with:
- Existing document or system
- Location
```

```
- Current owner
- Status: current, stale, duplicate, incomplete, missing, or needs review
- Evidence
- Required next action
- Whether professional review is required

Then build an execution-ready GTM operating system.

## Core commercial assumptions

Use these as planning assumptions, clearly labeled as assumptions rather than
guarantees or public commitments:

- Student Premium: $8.99/month or $69/year
- Institutional platform: $18 per enrolled student/year
- Institutional annual minimum: $30,000
- Implementation: $35,000-$150,000 one-time, based on complexity
- AI overage: $30 per 1,000 governed requests
- Premium support: 15% of annual platform fee, $15,000 minimum
- Optional marketplace/partner commission: 12%
- Institutional AI allocation: 2,400 pooled governed requests per enrolled
  student per year before overage

## First pilot requirement

Do not sell Semester initially as a total replacement for every university
system. Sell a paid pilot around one urgent institutional workflow.

Default proposed wedge:
“Semester Registration Readiness Pilot”

Pilot design:
- Buyer: Student Success, Advising, Registrar, or a defined academic division
- Scope: one cohort, department, or campus population
- Duration: 8-12 weeks or one academic term
- Outcome: improved readiness, action completion, clarity, and reduced support
  friction
- Required metric types:
  - activation
  - first meaningful action
  - workflow completion
  - weekly active usage
  - staff time/capacity
  - student confidence or satisfaction
  - support-volume or issue reduction
  - pilot-to-annual conversion readiness
- Commercial structure:
  - paid pilot
  - defined implementation scope
  - annual conversion price
  - named decision owner
  - agreed conversion date

## Required deliverables

Create or update all of the following:
```

```
### Strategy and positioning
- Ideal Customer Profile
- Initial segment and institutional beachhead
- Buyer and stakeholder map
- Category and positioning statement
- Messaging hierarchy
- Persona-specific value propositions
- Competitive battlecards
- Claims substantiation register
- Initial launch wedge and pilot offer

### Sales system
- 100-account target-list template
- Account scoring model
- Outreach sequences
- Discovery-call script
- Qualification scorecard
- Demo script
- Demo-environment checklist
- Pilot design-session agenda
- Pilot proposal template
- Mutual Action Plan
- Pricing calculator
- Quote/order-form intake checklist
- Procurement/security response pack
- Objection-handling library
- CRM stages, required fields, exit criteria, and forecast rules
- Weekly revenue review agenda
- Pilot-to-annual conversion playbook
- Expansion and reference-customer playbook

### Marketing system
- Marketing website information architecture
- Homepage and use-case-page copy requirements
- Trust Center content plan
- Demo request and lead-routing flow
- SEO topic and keyword map
- 90-day editorial calendar
- LinkedIn content plan for institutional buyers
- YouTube content plan for education/proof
- Instagram/TikTok plan for student adoption
- Webinar plan
- Email nurture sequence
- Retargeting plan
- Case-study and customer-proof system
- Press and partnership outreach plan

### Financial system
- 36-month driver-based financial model
- Conservative, base, ambitious, downside, delayed-sales, high-AI-use,
  pilot-heavy, and annual-contract-heavy scenarios
- Revenue, COGS, operating expense, cash, runway, CAC, LTV, payback,
  gross margin, retention, expansion, and break-even calculations
- Hiring plan tied to revenue/delivery capacity
- Budget and expense approval policy
```

```
- Monthly finance review format

### Customer success system
- Sales-to-CS handoff
- Welcome and kickoff sequence
- Implementation plan
- Customer RACI
- Data/integration readiness checklist
- Administrator and user training
- Campus launch kit
- Support process and severity model
- Customer health score
- Weekly pilot report
- Midpoint executive review
- Final value report
- QBR template
- Renewal/expansion playbook
- Churn-risk playbook
- Customer feedback taxonomy
- Case-study consent process

### Trust, security, and compliance
- Data inventory
- Data-flow diagrams
- Data classification
- Retention and deletion schedule
- Subprocessor register
- Security overview
- Privacy overview
- Incident response summary
- Business continuity and restore process
- Vendor risk review
- HECVAT Lite roadmap
- Security questionnaire response library
- Accessibility evidence roadmap
- Trust Center content inventory
- Compliance register with owner, evidence, status, renewal date, and risk

### Operations
- KPI tree
- Weekly operating calendar
- Decision log
- Risk register
- Executive dashboard
- Revenue, product, customer-success, finance, and security review cadence
- Quarterly OKR process
- Investor/board update template

## Rules

- Clearly distinguish assumptions, validated evidence, and future targets.
- Do not publish unverified product, security, accessibility, privacy,
  compliance, or customer-outcome claims.
- Legal, tax, accounting, insurance, privacy, security, and accessibility
  materials must be marked “requires professional review” where applicable.
- Do not rely on client-side UI logic as authorization.
```

```
- Do not use vanity metrics as proof of customer value.
- Do not recommend broad paid acquisition until the ICP, offer, landing page,
  conversion path, and sales follow-up process are functioning.
- Do not add duplicate docs when an existing artifact can be updated.

## Required final output

Return:
1. Existing versus missing asset inventory
2. P0, P1, and P2 GTM backlog
3. 30/60/90-day execution plan
4. Department owner/RACI matrix
5. Customer acquisition funnel and weekly targets
6. Financial model assumptions and scenario definitions
7. Pilot conversion plan
8. Trust/compliance evidence register
9. Required external systems and integrations
10. Exact next 10 actions in order
```

Interactive financial model

The report includes the model specification. Use the prompt below to build it as a React/Vite module in the existing Semester app or as a standalone internal tool first.

```
Build an interactive React financial model named
“Semester GTM Financial Model.”

Purpose:
Model the financial impact of institutional pilot sales, annual-license
conversion, student subscriptions, implementation revenue, AI usage, support,
customer acquisition, staffing, operating expenses, cash burn, runway, and
break-even.

Use local sample data only in the first version. The tool must export scenario
data as CSV and Markdown. Do not require a backend for the first version.

Model horizon:
- Monthly for 36 months
- Annual summary for Years 1, 2, and 3

Scenarios:
- Conservative
- Base
- Ambitious
- Downside / delayed sales
- Pilot-heavy
- Annual-contract-heavy
- High AI usage
- Low student adoption
- High implementation cost
- Long procurement cycle

Editable assumptions:
```

```
Pricing:
- Student Premium monthly price: $8.99
- Student Premium annual price: $69
- Institution price per enrolled student/year: $18
- Institution annual minimum: $30,000
- Implementation fee range: $35,000-$150,000
- AI overage price: $30 per 1,000 requests
- Premium support: 15% of annual platform fee, $15,000 minimum
- Marketplace commission: 12%, optional

Funnel:
- Target accounts
- Contact rate
- Reply rate
- Discovery rate
- Qualification rate
- Demo rate
- Pilot-design rate
- Proposal rate
- Pilot close rate
- Pilot activation rate
- Pilot success rate
- Pilot-to-annual conversion rate
- Annual expansion rate
- Sales-cycle length in months
- Payment collection timing

Customer:
- Institution count
- Average enrolled students per institution
- Student invite rate
- Student activation rate
- Student Premium conversion rate
- Monthly/annual student churn
- Institutional gross retention
- Institutional expansion rate

Costs:
- Cloud cost per active user
- Storage cost per active user
- AI cost per request
- Support cost per customer
- Customer-success cost per pilot
- Implementation labor cost
- Sales payroll
- Marketing spend
- Engineering payroll
- Contractor spend
- Legal/compliance cost
- Accounting/tax cost
- Insurance cost
- Software/vendor cost
- Travel/events cost
- General/administrative cost

Required calculations:
```

```
- Monthly recurring revenue
- Annual recurring revenue
- Pilot revenue
- Implementation revenue
- Premium support revenue
- AI overage revenue
- Student subscription revenue
- Marketplace/partner revenue
- COGS
- Gross profit
- Gross margin
- Operating expenses
- EBITDA-style operating result
- Cash burn
- Ending cash
- Runway
- CAC
- CAC payback months
- LTV sensitivity
- Net revenue retention
- Break-even month
- Revenue per employee
- Implementation capacity constraint

Required UI:
- Scenario dropdown
- Editable numeric inputs and sliders
- Assumption reset button
- Monthly revenue table
- Monthly expense table
- Cash/runway table
- Funnel visualization
- Break-even chart
- Revenue mix chart
- CAC/LTV summary tiles
- Sensitivity table for pilot conversion versus sales cycle length
- Sensitivity table for AI cost versus AI overage pricing
- Warnings when:
  - gross margin falls below threshold
  - cash runway falls below 12 months
  - CAC payback exceeds target
  - implementation demand exceeds team capacity
  - AI costs exceed included revenue
- Export current scenario to CSV
- Export current scenario to Markdown
- Copy assumptions as a prompt

Formula requirements:
- CAC = Sales and marketing spend / new customers acquired
- CAC payback months = CAC / monthly gross profit per new customer
- LTV = annual recurring revenue × gross margin / annual logo churn
- MRR = recurring annual contract value / 12 plus recurring student revenue
- Break-even = first month cumulative gross profit covers cumulative operating expens
- Clearly label all outputs as forecast estimates driven by assumptions.

Design:
```

```
- Premium B2B finance dashboard
- Accessible controls and table navigation
- Clear distinction between inputs, calculated outputs, and warnings
- Use US dollars
- No external chart library required; use simple CSS/SVG/chart primitives if needed.
```

Financial-model layout

Tab	Purpose
Assumptions	All editable commercial, funnel, cost, and hiring inputs
Funnel	Account volume and stage conversion rates
Revenue	Pilot, implementation, annual, support, AI, student, and partner revenue
Costs	COGS, payroll, vendors, marketing, legal, insurance, and overhead
Cash	Burn, collections, runway, funding need, and break-even
Unit economics	CAC, payback, LTV range, margin, retention, and expansion
Scenarios	Conservative, base, ambitious, downside, and operating sensitivities
Exports	CSV, Markdown, scenario snapshot

Pilot sales email sequence

Use this only with highly relevant, researched contacts. Each message needs an actual institutional reason for outreach; do not send generic mass email.

Email 1 — Problem-led introduction

Subject options

- Reducing registration friction for [Institution] students
- A practical registration-readiness pilot for [Institution]
- Question about student planning before registration
- [Name], a small student-success pilot idea

```
Hi [First Name],

I’m Harrison, building Semester—a student operating system designed to help
institutions turn fragmented student planning and support workflows into clear,
trackable actions.

I reached out because [personalized institutional trigger, initiative, program,
or public challenge].

We are speaking with [role/team] leaders about a narrowly scoped Registration
Readiness Pilot. The goal is not to replace existing systems. It is to help a
defined student cohort understand next steps, complete priority actions, and
give staff a clearer view of where students are getting stuck.
```

```
For a pilot, we would jointly measure:
- student activation
- first meaningful action
- readiness/workflow completion
- support or advising friction
- student confidence and satisfaction

Would a 20-minute conversation next week be useful to compare your current
process and see whether this is relevant for [Institution]?

Best,
Harrison Rubin
Founder, Semester
[Website]
[Calendar link]
```

Email 2 — Follow-up with concrete value

Send: 3–5 business days after Email 1.

Subject options

- Re: registration readiness at [Institution]
- One measurable pilot, not another platform replacement
- A low-risk way to test student workflow improvement

```
Hi [First Name],

Following up because the important part of the idea is its scope:

Semester would begin with one cohort, one workflow, and a fixed pilot period—
not a broad system replacement.

A typical pilot includes:
- a jointly chosen student cohort
- agreed baseline and outcome metrics
- a configured student action workflow
- admin/champion training
- weekly operating review
- midpoint executive review
- final value report and annual conversion decision

If [Institution] is already working on [specific initiative], I can send a
one-page pilot outline tailored to that context.

Is [day/time option] or [day/time option] easier for a short conversation?

Best,
Harrison
```

Email 3 — Proof and procurement readiness

Send: 5–7 business days after Email 2.

Subject options

- Pilot structure, data boundaries, and institutional review
- How Semester handles a scoped institutional pilot
- Security and implementation questions for a Semester pilot

```
Hi [First Name],

One common question is whether a pilot creates a large IT, security, or
procurement burden.

We structure the initial deployment around:
- limited scope and a defined cohort
- explicit data and integration assumptions
- named customer and Semester owners
- a documented implementation plan
- security/privacy documentation appropriate to the scope
- clear success metrics and an annual conversion decision

We are building the pilot so your team can evaluate a real outcome without
committing to a broad multi-year rollout upfront.

If the timing is not right, I'm also happy to send the pilot framework for
future planning. Should I send that, or is there a better colleague to speak
with?

Best,
Harrison
```

Email 4 — Breakup / referral request

Send: 7–10 business days after Email 3.

Subject: Close the loop?

```
Hi [First Name],

I have not heard back, so I will close the loop after this note.

Semester may not be relevant for [Institution] right now. If improving student
readiness, advising workflows, or registration action completion is on the
roadmap, I would be glad to reconnect when timing is better.

If there is another person who owns [student success / advising / registration
readiness / student experience], I would appreciate a referral.

Thank you,
Harrison
```

After a discovery call

Subject: Next steps: Semester + [Institution] pilot design

```
Hi [First Name],

Thank you for the conversation today.

My understanding of the current opportunity:
- Current workflow: [summary]
- Target student population: [summary]
- Primary friction: [summary]
- Existing tools/process: [summary]
- Desired outcome: [summary]
- Potential success metrics: [summary]
- Stakeholders: [summary]
- Decision timing: [summary]

Proposed next step:
A 45-minute pilot design session with [champion], [technical/IT contact], and
[decision-maker if possible] to agree on scope, baseline, data assumptions,
implementation timeline, and a conversion decision.

I will bring a draft pilot outline and Mutual Action Plan.

Best,
Harrison
```

Pilot proposal follow-up

Subject: Semester pilot proposal and Mutual Action Plan

```
Hi [First Name],

Attached is the proposed Semester pilot plan for [Institution].

It includes:
- the defined workflow and cohort
- roles and responsibilities
- data/integration assumptions
- implementation timeline
- training and support plan
- success metrics and baseline
- weekly review cadence
- pilot fee and annual conversion path
- proposed conversion decision date

The main open items are:
1. [Open item]
2. [Open item]
3. [Open item]

Could we schedule a review with the appropriate operational, IT/security, and
executive stakeholders to finalize the scope?
```

```
Best,
Harrison
```

Cloud data-security plan

Use this as the minimum operational baseline. The FTC's guidance supports an approach centered on data minimization, reasonable safeguards, secure disposal, and a planned response to security incidents.[4] [5] [6]

Security operating principles

```
1. Collect only data required for a defined product workflow.
2. Classify data before storing or sharing it.
3. Default to least privilege and tenant-scoped access.
4. Keep privileged credentials server-only.
5. Encrypt data in transit and use provider-supported encryption at rest.
6. Log sensitive administration and data-access events.
7. Test restoration, not only backup creation.
8. Review vendors before sharing customer data.
9. Practice incident response before a real incident.
10. Do not make security or compliance claims without evidence.
```

Security plan checklist

Domain	Control	Evidence to retain	Owner
Governance	Security owner, policy calendar, risk register	Charter, review log	Founder/security lead
Data inventory	Data fields, source, purpose, owner, retention	Data inventory + diagrams	Privacy/product
Access	MFA, role permissions, quarterly review, offboarding	Access review logs	Engineering/security
Secrets	Secret manager, scanning, rotation process	Scan reports, secret inventory	Engineering
App security	PR review, dependency review, security release gate	CI logs, PR records	Engineering
Data protection	TLS, encryption configuration, minimization	Architecture/config evidence	Engineering
Tenant isolation	RLS, org/user scopes, server-side authorization	Migration/policy tests	Engineering
Logging	Error tracking, audit events, alert triage	Dashboard/runbook	Engineering/ops
Backups	Backup policy, restore procedure, restore test	Restore-test record	Engineering
Incident response	Severity model, contacts, templates, tabletop	IR plan/exercise notes	Security/ops
Vendor risk	Subprocessor list, review process, contracts	Vendor register	Ops/security

Domain	Control	Evidence to retain	Owner
Privacy	Retention/deletion, request handling, policy review	Deletion logs/policy	Privacy/legal
Procurement	HECVAT Lite, questionnaire library, DPA path	Evidence register	Sales/security
Accessibility	Baseline testing, issue tracking, statement	Test results/backlog	Product/design

Incident severity model

Severity	Example	Initial target	Customer communication
P0	Confirmed unauthorized access, material data exposure, total service outage	Immediate internal escalation	Counsel/security-approved notice process
P1	Critical workflow unavailable for a meaningful customer segment	Rapid triage and regular updates	Affected customers updated on defined cadence
P2	Major feature degraded; workaround exists	Same business day triage	Customer support response as appropriate
P3	Minor defect, request, or documentation issue	Planned prioritization	Standard support process

HECVAT readiness plan

1. Download the current HECVAT template from EDUCAUSE.
2. Identify which required questions apply to Semester.
3. Map each answer to a control owner and evidence file.
4. Never answer “yes” without evidence.
5. Keep evidence versioned and dated.
6. Use HECVAT Lite first for lower-risk/scoped pilot review where accepted.
7. Prepare Full HECVAT evidence as target institutions require it.
8. Keep security, privacy, architecture, accessibility, subprocessors, incident response, and data-retention documents synchronized.

EDUCAUSE provides instructions for completing HECVAT and notes that the vendor should respond to the required questions applicable to its services. [7] [1]

PDF report

The **Semester GTM Playbook PDF report** generated in the prior response contains the complete packaged version of this material, including:

- GTM and pilot structure
- Pricing assumptions
- 36-month financial-model requirements
- Sales funnel
- Customer success/onboarding system

- Pilot agreement business outline
- Security and HECVAT readiness plan
- 90-day execution sequence
- Operating cadence
- Master execution prompt

⁂

1. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/how-to-use-the-higher-education-community-vendor-assessment-toolkit
2. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/hecvat-faqs-for-higher-education
3. https://er.educause.edu/articles/2025/2/hecvat-4-better-than-ever
4. https://consumer.ftc.gov/business-guidance/privacy-security/data-security
5. https://www.ftc.gov/business-guidance/blog/2012/01/collection-protection
6. https://www.ftc.gov/sites/default/files/documents/public_statements/prepared-statement-federal-trade-commission-data-security/110615datasecurityhouse.pdf
7. https://www.ren-isac.net/services/hecvat.html

perplexity

Draft the master GTM playbook prompt

Generate the interactive financial model
Write the pilot sales email sequence
Build the compliance data security plan
Export this playbook as a PDF report

I created the **Semester GTM Playbook PDF report**. It includes the master GTM prompt, financial-model specification, institutional pilot sales playbook, customer-success workflow, acquisition funnel, pricing assumptions, and cloud security/compliance checklist.

The report is a business-operating draft: use qualified counsel, accountants, insurers, and security reviewers before relying on contracts, privacy commitments, tax treatment, insurance limits, accessibility claims, or institutional compliance representations.

What's in the PDF

The PDF includes:

- **Launch wedge:** a Registration Readiness Pilot model, scoped to one cohort, department, or institutional workflow.
- **Commercial assumptions:** Student Premium, institutional annual licensing, implementation fees, AI overages, premium support, and optional partner revenue.
- **Financial model design:** a driver-based 36-month model with conservative, base, ambitious, downside, delayed-sales, high-AI-use, pilot-heavy, and annual-contract-heavy scenarios.
- **Customer acquisition funnel:** account targeting → discovery → demo → pilot design → security/procurement → pilot → annual conversion → expansion.
- **Sales playbook:** discovery questions, pilot proposal structure, Mutual Action Plan, CRM requirements, and weekly scorecard.
- **Pilot agreement business outline:** essential terms for counsel to convert into an enforceable agreement.
- **Customer success lifecycle:** sales handoff, kickoff, configuration, training, launch, adoption, midpoint review, value report, renewal, and expansion.
- **Cloud security/compliance plan:** identity, secrets, secure development, data protection, monitoring, backups, incident response, vendor review, accessibility, and HECVAT readiness.
- **90-day execution sequence** and operating cadence.

- A copy-ready **master GTM execution prompt**.

Semester's pricing assumptions in the report are framed as planning inputs—$8.99/month or $69/year for Student Premium; $18 per enrolled student/year with a $30,000 annual institutional minimum; $35,000–$150,000 implementation; $30 per 1,000 AI-request overage; and premium support at 15% of annual platform fees with a $15,000 minimum. These should be tested against delivery costs, buyer feedback, and actual margin before being treated as a public price card.

Customer acquisition funnel format

Use this as the operating funnel inside your CRM:

```
100 Target Accounts
  ↓
40–60 Personalized Contacts
  ↓
12–20 Replies / Referrals
  ↓
8–15 Discovery Meetings
  ↓
5–10 Qualified Opportunities
  ↓
4–8 Role-Specific Demos
  ↓
3–5 Pilot Design Sessions
  ↓
2–4 Security / Procurement Reviews
  ↓
2–3 Pilot Proposals
  ↓
1–3 Signed Paid Pilots
  ↓
Pilot Activation and KPI Attainment
  ↓
Annual Conversion
  ↓
Expansion, Renewal, Case Study, Reference
```

Track every account using:

Funnel stage	Required evidence	Core metric
Target account	Buyer hypothesis, trigger event, priority score	Account coverage
Discovery	Problem, champion, budget, timeline, existing process	Qualification rate
Demo	Attendees, workflow shown, objections, next step	Demo-to-design rate
Pilot design	Scope, cohort, metrics, integration assumptions	Proposal readiness
Security/procurement	HECVAT/DPA/security owner/legal path	Review progress
Pilot proposal	Price, scope, conversion date, Mutual Action Plan	Pilot close rate
Activated pilot	Invited users, configured admin, baseline metrics	Activation rate

Funnel stage	Required evidence	Core metric
Success achieved	Workflow completion, outcome report	Pilot KPI attainment
Annual conversion	Order form, annual scope, close date	Pilot-to-annual rate

Institutional pilot agreement outline

Use this with counsel as the structural brief—not as unsigned legal advice:

```
1. Parties and effective date
2. Pilot purpose and defined Semester workflow
3. Pilot scope, included users/cohort, exclusions, and non-goals
4. Pilot term, implementation milestones, and termination rights
5. Institution responsibilities:
   - executive sponsor
   - operational champion
   - data/access approvals
   - stakeholder participation
   - communications and training support
6. Semester responsibilities:
   - configuration
   - implementation support
   - documented support path
   - reporting cadence
   - verified security commitments
7. Fees, invoicing, payment terms, taxes, and implementation fees
8. Success metrics, baseline, reporting, midpoint review, and final review
9. Data protection:
   - data categories
   - permitted processing
   - retention
   - deletion/return
   - DPA and subprocessors
10. Security, accessibility, support, and incident commitments
11. Confidentiality, IP, feedback, and publicity/reference rights
12. Warranty/disclaimer, liability, indemnity, governing law
13. Annual conversion:
   - annual price
   - conversion date
   - decision owner
   - conversion criteria
   - expansion options
```

Cloud security checklist

FTC guidance emphasizes collecting only what is needed, protecting retained data, disposing of it when no longer necessary, and maintaining incident-response readiness.[1] [2] [3]

Domain	Minimum evidence before pilot
Governance	Security owner, risk register, policy review cycle
Identity	MFA, least privilege, offboarding process

Domain	Minimum evidence before pilot
Secrets	Managed secrets, secret scan, no browser-exposed privileged keys
Data inventory	Fields, sources, purposes, owners, vendors, retention, deletion
Secure development	PR review, dependency scanning, release checklist, vulnerability process
Data protection	TLS, provider-backed encryption at rest, data minimization
Monitoring	Error monitoring, audit events, alert triage
Backups	Backup scope, recovery procedure, restore-test record
Incident response	Severity matrix, contacts, customer templates, tabletop exercise
Vendor risk	Subprocessor register and review process
Higher-ed procurement	HECVAT Lite plan, security questionnaire library, DPA path
Accessibility	Public statement, known-gap log, remediation process

For university procurement, HECVAT is a key preparation artifact: EDUCAUSE describes it as a comprehensive framework for evaluating technology vendors' security, privacy, accessibility, and related risk controls.[4] [5] [6]

Recommended next sequence

1. Choose the exact pilot wedge: buyer, cohort, workflow, baseline, and outcome.
2. Build the financial model using the PDF's driver structure before finalizing public pricing.
3. Set up CRM stages and load 100 target institutional accounts.
4. Finalize the pilot proposal, Mutual Action Plan, sales deck, and demo.
5. Complete the initial Trust Center and HECVAT Lite evidence roadmap.
6. Have counsel review the pilot agreement, MSA, DPA, Terms, Privacy Policy, and SLA.
7. Implement the customer onboarding, health score, support, and renewal workflow before the first signed pilot.
8. Measure activation, first meaningful action, workflow completion, support friction, and conversion—not only downloads or social engagement.

⁂

1. https://consumer.ftc.gov/business-guidance/privacy-security/data-security
2. https://www.ftc.gov/business-guidance/blog/2012/01/collection-protection
3. https://www.ftc.gov/sites/default/files/documents/public_statements/prepared-statement-federal-trade-commission-data-security/110615datasecurityhouse.pdf
4. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/how-to-use-the-higher-education-community-vendor-assessment-toolkit
5. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/hecvat-faqs-for-higher-education

6. https://er.educause.edu/articles/2025/2/hecvat-4-better-than-ever
7. https://www.perplexity.ai/search/4cab66a0-68e7-4697-b060-725e6bb37917
8. https://www.sba.gov/counseling/launch-your-business/
9. https://www.irs.gov/businesses/small-businesses-self-employed/checklist-for-starting-a-business
10. https://www.sba.gov/counseling/
11. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit
12. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/hecvat-faqs-for-corporations
13. https://aws.amazon.com/blogs/publicsector/higher-education-community-vendor-assessment-toolkit-now-available-on-aws-artifact/
14. https://www.ren-isac.net/services/hecvat.html
15. https://wolfia.com/hecvat
16. https://www.sba.gov/counseling/plan-your-business/
17. https://www.manatt.com/insights/articles/2016/new-ftc-guide-offers-tips-for-protecting-personal
18. https://www.perplexity.ai/search/a52de1ab-276c-4fc0-b475-dc0dff5fee6b

Show the finance and accounting operations section

Explain the Go-to-Market and Revenue operations
Show the security and compliance systems
Explain the Customer Success and Implementation side
Show the People hiring and culture systems

Semester needs five business operating systems that work together: **finance/accounting**, go-to-market/revenue, security/compliance, customer success/implementation, and people/culture. These are not back-office extras—they are what make a large Education OS credible, sellable, supportable, and durable.

The operating rule is:

> Every customer promise must connect to a financial owner, delivery owner, support process, security/privacy control, product capability, and evidence trail.

Finance and accounting operations

Semester's finance system must answer, at any time:

```
How much cash do we have?
How long is the runway?
What is committed revenue?
What is actually collected?
Which contracts/pilots are profitable?
What does each customer cost to serve?
What is the margin impact of AI and infrastructure?
What can we afford to hire or build next?
```

Finance architecture

```
Finance and Accounting OS
├── Chart of accounts
├── Bank and cash management
├── Accounts receivable
├── Accounts payable
├── Billing
├── Invoices
├── Payments
├── Collections
├── Revenue recognition review
├── Pricing and discounts
├── Budget
```

```
├── Forecast
├── Cash flow
├── Payroll and contractors
├── Expense management
├── Tax calendar
├── Insurance
├── Vendor spend
├── Unit economics
├── Customer profitability
├── AI/cloud cost allocation
├── Financial controls
├── Monthly close
├── Audit support
├── Board reporting
└── Fundraising model
```

Finance screens and console modules

```
/ops/finance
├── Cash dashboard
├── Runway dashboard
├── Revenue dashboard
├── Revenue forecast
├── P&L view
├── Balance sheet view
├── Cash-flow view
├── Budget vs actual
├── Expense categories
├── Vendor spend
├── Payroll and contractor forecast
├── Unit economics
├── Customer profitability
├── AI cost dashboard
├── Cloud cost dashboard
├── Pricing calculator
├── Discount approvals
├── Finance risk alerts
├── Collections/dunning
├── Invoices
├── Payment reconciliation
├── Tax calendar
├── Insurance tracker
├── Board finance pack
└── Forecast scenarios
```

Core financial data model

```
Finance
├── Accounting periods
├── Accounts
├── Journal entry references
├── Bank accounts
├── Cash snapshots
```

```
├── Budgets
├── Forecast scenarios
├── Revenue plans
├── Expense plans
├── Cost centers
├── Departments
├── Customers
├── Contracts
├── Orders
├── Subscriptions
├── Invoices
├── Invoice lines
├── Payments
├── Credits/refunds
├── Dunning cases
├── Vendor contracts
├── Purchase orders
├── Expenses
├── Payroll costs
├── Contractor costs
├── Cloud costs
├── AI costs
├── Security/compliance costs
└── Financial reporting snapshots
```

The product database should track commercial and operational billing state—quotes, orders, subscriptions, invoices, entitlements, payment events, dunning, and customer profitability. It should **not** become your statutory accounting ledger without accountant-approved design and a proper accounting system.

Revenue model

```
Revenue
├── Paid institutional pilots
├── Implementation fees
├── Annual institutional platform licenses
├── Per-enrolled-student license fees
├── Premium support
├── AI usage overages
├── Student Premium subscriptions
├── Department plans
├── Enterprise/multi-campus plans
├── Marketplace commissions
├── Partner revenue
├── Professional services
├── Training services
└── Data migration/integration services
```

Planning prices

```
Student Premium:
$8.99/month or $69/year

Institution Platform:
$18/enrolled student/year
$30,000 annual minimum

AI overage:
$30 per 1,000 governed requests

Implementation:
$35,000-$150,000 one-time

Premium support:
15% of annual platform fee
$15,000 minimum

Marketplace/partner commission:
12%, optional and separate from core viability
```

Finance workflow

```
Quote created
→ pricing exception check
→ approval if required
→ customer accepts order form
→ subscription/contract activated
→ invoice generated
→ payment collected/reconciled
→ entitlement granted
→ revenue recognized/reviewed
→ renewal tracked
→ expansion/cancellation/dunning handled
→ customer profitability reviewed
```

Financial controls

Action	Required control
Create draft quote	Revenue operations role
Approve discount	Deal desk based on margin/ACV threshold
Send final quote	Revenue approval
Change commercial plan	Approval plus entitlement impact review
Issue credit/refund	Finance approval plus invoice/payment reference
Write off receivable	Finance + executive approval above threshold
Change billing account	Verified customer authority

Action	Required control
Change bank/payment settings	Strong authentication + finance approval
Publish financial forecast	Finance owner review
Board/investor reporting	Founder/finance approval
Create vendor commitment	Procurement/budget owner approval

Financial KPIs

```
Cash on hand
Monthly burn
Runway
Committed ARR
Recognized revenue
Collected cash
MRR
ARR
Pilot revenue
Implementation revenue
Expansion revenue
Gross margin
Contribution margin
AI cost per active user
Cloud cost per active user
Support cost per customer
Implementation cost per pilot
CAC
CAC payback
LTV sensitivity
Gross retention
Net revenue retention
Average contract value
Sales cycle
Invoice aging
Collections rate
Dunning recovery
Budget variance
Headcount cost
Revenue per employee
```

Monthly close

```
Close calendar opens
→ invoices reviewed
→ payments reconciled
→ vendor invoices recorded
→ payroll and contractor costs recorded
→ cloud and AI costs allocated
→ expenses categorized
→ deferred revenue reviewed
→ P&L generated
```

```
→ balance sheet reviewed
→ cash flow updated
→ forecast variance explained
→ runway updated
→ executive review
→ board/investor summary
```

Go-to-Market and Revenue Operations

Semester's GTM system has two connected engines:

```
Institutional B2B engine:
Paid pilots → annual contracts → expansion → reference customers

Student adoption engine:
Student value → activation → retained use → institutional proof → referrals
```

GTM architecture

```
GTM and Revenue OS
├── Market definition
├── ICP
├── Segmentation
├── Account scoring
├── Buyer mapping
├── CRM
├── Outreach
├── Discovery
├── Qualification
├── Demos
├── Pilot design
├── Security/procurement
├── Proposal
├── Mutual Action Plan
├── Contracting
├── Implementation handoff
├── Pipeline forecasting
├── Sales enablement
├── Marketing automation
├── Content and demand generation
├── Channel partners
├── Customer proof
├── Renewal/expansion
├── Attribution
├── Pricing governance
└── Revenue analytics
```

Sales funnel

```
Target Accounts
  ↓
Researched Accounts
  ↓
Personalized Outreach
  ↓
Replies / Referrals
  ↓
Discovery Meetings
  ↓
Qualified Opportunities
  ↓
Role-Specific Demos
  ↓
Pilot Design Sessions
  ↓
Security / Privacy / Procurement Review
  ↓
Pilot Proposal
  ↓
Paid Pilot Signed
  ↓
Tenant Implementation
  ↓
Pilot Activation
  ↓
Measured Success
  ↓
Annual Contract Conversion
  ↓
Expansion / Renewal / Reference
```

CRM data requirements

```
Account
├── Institution name
├── Segment
├── Enrollment range
├── System type
├── Geography
├── Existing technology stack
├── Priority score
├── Trigger event
├── Champion
├── Economic buyer
├── Technical buyer
├── Security/privacy reviewer
├── Procurement/legal contact
├── Current stage
├── Estimated ACV
├── Pilot value
├── Annual potential
```

```
├── Probability
├── Close date
├── Risks
├── Next step
└── Owner

Opportunity
├── Problem statement
├── Current workflow
├── Scope
├── Student cohort
├── Buyer goals
├── Baseline metrics
├── Success metrics
├── Security requirements
├── Integration needs
├── Commercial terms
├── Mutual Action Plan
├── Decision process
├── Competitors
├── Objections
└── Win/loss reason
```

Sales stages

Stage	Exit criteria
Target	Account researched and scored
Contacted	Relevant stakeholder reached with personalized message
Discovery	Pain, champion, timeline, and buyer path documented
Qualified	Problem, budget path, decision process, and deployment feasibility confirmed
Demo	Use case demonstrated to required stakeholders
Pilot design	Cohort, scope, baseline, success criteria, and implementation assumptions agreed
Procurement/security	Review owners, artifacts, and decision path identified
Proposal	Price, terms, annual conversion path, and decision date delivered
Contracting	Legal/procurement actively reviewing agreement
Closed won: pilot	Agreement executed; implementation kickoff scheduled
Pilot active	Cohort launched; success metrics measured
Conversion	Annual decision in process
Closed won: annual	Annual agreement active
Expansion	Additional campus, department, module, or user scope identified
Closed lost	Reason, competitor, and future re-engagement path recorded

Required sales assets

```
ICP document
Buyer persona library
Account scoring model
Discovery script
Qualification scorecard
Demo script
Demo environment
Pilot design agenda
Pilot proposal
Mutual Action Plan
ROI calculator
Pricing calculator
Quote template
Order form intake
Security overview
HECVAT roadmap
DPA/SLA process
Accessibility response
Procurement FAQ
Competitive battlecards
Objection-handling library
Case-study template
Reference-call guide
Renewal/expansion deck
```

Marketing and acquisition infrastructure

```
Website
├── Homepage
├── Product pages
├── Pilot page
├── Pricing
├── Demo request
├── Trust Center
├── Resource hub
├── SEO pages
├── Case studies
└── Contact

Content
├── LinkedIn
├── YouTube
├── Webinars
├── Newsletter
├── Guides
├── Research
├── Founder content
├── Student content
└── Partner content

Lifecycle
├── Lead nurture
```

```
├── Demo follow-up
├── Webinar follow-up
├── Pilot nurture
├── Customer onboarding
├── Student activation
├── Retention
├── Renewal
└── Referral
```

Security and compliance systems

Security and compliance must be treated as continuous operational systems—not a “launch checklist.”

Security architecture

```
Security Program
├── Governance
├── Asset inventory
├── Data inventory
├── Data classification
├── Threat modeling
├── Identity and access
├── MFA
├── Secrets management
├── Secure development lifecycle
├── Dependency management
├── Vulnerability management
├── Logging and monitoring
├── Incident response
├── Backup and restore
├── Disaster recovery
├── Business continuity
├── Vendor risk
├── Access reviews
├── Penetration testing
├── Security awareness
├── Evidence register
├── Trust Center
├── HECVAT
├── Security questionnaires
└── Security reporting
```

Security operations workflow

```
Asset/data inventory
→ risk assessment
→ control design
→ implementation
→ evidence collection
→ testing
→ monitoring
```

```
→ alert or incident
→ triage
→ remediation
→ verification
→ control review
→ customer/procurement evidence
```

Security controls

Domain	Required controls
Identity	MFA, least privilege, joiner/mover/leaver, service-account ownership
Authorization	RLS, tenant isolation, capability checks, server-side scope enforcement
Secrets	Server-only credentials, rotation, secret scanning, no browser exposure
Application	Code review, testing, SAST, dependency scanning, release gates
Infrastructure	Environment separation, logging, monitoring, backups, recovery
Data	Classification, minimization, encryption, retention, deletion, export
AI	Provider controls, policy, classification, audit, evaluation, spend limits
Vendors	Security review, contract terms, subprocessors, monitoring, offboarding
Incident	Severity, command roles, communications, postmortem, corrective actions
Compliance	HECVAT, evidence, control mapping, review calendar, claims governance

Privacy system

```
Privacy Operations
├── Data inventory
├── Data map
├── Classification
├── Purpose limitation
├── Consent
├── Retention
├── Deletion
├── Legal holds
├── Subject requests
├── Export requests
├── Erasure requests
├── Correction requests
├── Subprocessors
├── Privacy impact assessments
├── Privacy incidents
├── Policy review
└── Privacy reporting
```

Compliance evidence record

```
Control
→ owner
→ policy
→ implementation
→ evidence
→ test
→ reviewer
→ last review
→ next review
→ customer claim
→ procurement response
→ exception/remediation
```

Customer Success and Implementation

The customer-success system converts a signed contract into activation, proof, renewal, and expansion.

Customer lifecycle

```
Contract signed
→ sales-to-CS handoff
→ kickoff
→ implementation plan
→ data/integration readiness
→ tenant configuration
→ admin training
→ student/staff launch
→ activation
→ adoption
→ weekly operating review
→ midpoint executive review
→ final value report
→ annual conversion
→ QBR
→ renewal
→ expansion
→ reference/case study
```

Sales-to-CS handoff

Sales must deliver:

```
Signed agreement
Commercial scope
Implementation scope
Pilot timeline
Annual conversion date
Executive sponsor
```

```
Economic buyer
Operational champion
Technical/IT contact
Security/privacy/procurement contacts
Target cohort
Workflow
Baseline metrics
Success criteria
Promised integrations
Data categories
Known risks
Customer commitments
Semester commitments
Decision process
```

Implementation workspace

```
Implementation Project
├── Project plan
├── RACI
├── Milestones
├── Dependencies
├── Data readiness
├── Integration readiness
├── Tenant setup
├── Role setup
├── Training
├── Launch communications
├── Risk register
├── Issue tracker
├── Status report
├── Acceptance testing
├── Go-live gate
└── Transition to success
```

Customer health score

```
Customer Health
├── Commercial
├── Implementation
├── Activation
├── Adoption
├── Workflow completion
├── Support sentiment
├── Executive engagement
├── Renewal status
├── Integration health
├── Security/compliance status
├── Product fit
└── Expansion signal
```

Example weights

Signal	Weight
Admin engagement	15%
Student/user activation	20%
First meaningful action	15%
Core workflow completion	20%
Weekly active use	10%
Support sentiment	5%
Executive engagement	5%
Pilot/renewal outcome trend	10%

Customer-success assets

```
Welcome sequence
Kickoff deck
Implementation plan
RACI
Data readiness checklist
Admin setup guide
Training curriculum
Office hours plan
Student launch kit
Support policy
Escalation matrix
Weekly pilot report
Midpoint executive review
Final value report
QBR
Renewal plan
Expansion plan
Churn-risk playbook
Case-study consent
Reference process
Customer advisory board charter
```

People, hiring, and culture systems

Semester's product ambition requires a company that can operate responsibly under complexity.

People operating system

```
People OS
├── Org design
├── Workforce plan
├── Hiring plan
├── Role scorecards
├── Recruiting
├── Interview loops
├── Reference checks
├── Contractor management
├── Onboarding
├── Access provisioning
├── Security training
├── Performance management
├── Compensation
├── Equity
├── Benefits
├── Career development
├── Learning and development
├── Culture
├── Employee handbook
├── Incident/ethics reporting
├── Offboarding
├── Access revocation
└── Alumni network
```

Org design

```
Founder / CEO
├── Product and Design
├── Engineering and Platform
├── Learning and Academic Operations
├── Student Success and Implementation
├── Sales and GTM
├── Customer Success and Support
├── Security, Privacy, Compliance, and Legal Ops
├── Finance and Business Operations
├── Partnerships and Developer Ecosystem
└── People and Culture
```

Hiring sequence

Stage	First priority hires
Prototype to pilot	Full-stack/platform engineer, product designer, implementation/customer-success lead
First paid pilots	Institutional sales lead, security/compliance adviser, support/implementation capacity
Pilot conversion	Product manager, data/analytics, academic operations/registrar expert
Expansion	Platform/SRE, partnerships/integrations, marketing/content, finance/operations
Platform scale	Developer relations, marketplace/partner team, regional operations, research/outcomes

Hiring scorecard

Every role should have:

```
Mission
Outcomes
Core competencies
Technical/domain requirements
Decision rights
Metrics
Interview process
Reference checks
Compensation range
Equity range
Onboarding plan
30/60/90-day deliverables
Access needed
Security/privacy training
Performance-review expectations
```

Cultural standards

```
Student agency over surveillance
Evidence over hype
Accessibility by default
Trust before shortcuts
Interoperability over lock-in
Human judgment in high-stakes decisions
Reliability before vanity features
Documentation is part of the product
Every critical workflow has a rollback
Every public claim has evidence
Every customer promise has an owner
Every incident creates a learning loop
```

Integrated business operating cadence

Cadence	Review	Required output
Daily	Operations triage	P0/P1 issues, customer blockers, incident decisions
Weekly	Revenue review	Pipeline, deals, pilots, forecast, next steps
Weekly	Product/engineering	Releases, quality, technical debt, customer feedback
Weekly	Customer success	Adoption, support, health, renewal/pilot risk
Weekly	Security/reliability	Vulnerabilities, incidents, SLOs, access, changes
Weekly	Marketing/growth	Funnel, content, campaigns, activation, attribution
Monthly	Finance	P&L, cash, burn, runway, budget variance, collections
Monthly	Trust/compliance	Evidence, reviews, vendor risk, HECVAT, privacy

Cadence	Review	Required output
Monthly	Leadership	KPI review, strategic decisions, hiring, risks
Quarterly	OKR/board	Strategy, roadmap, capital, performance, risk, priorities

Master business execution command

```
Build the complete Semester Business and Operations System.

Semester is an Education Operating System and Company of Education. It requires
the same level of operational coherence as the product itself.

Create and operationalize five connected systems:

1. Finance and Accounting Operations
2. Go-to-Market and Revenue Operations
3. Security, Privacy, Accessibility, and Compliance
4. Customer Success and Implementation
5. People, Hiring, and Culture

For each system create:

- Operating model
- Owner roles
- Workflows
- Console screens
- Systems of record
- Data model
- Required integrations
- KPIs
- Dashboards
- Meeting cadence
- Approval thresholds
- Risk controls
- Required documents
- Required templates
- Customer impact
- Product dependencies
- Security/privacy implications
- Budget/capacity model
- Current-state audit
- Gap register
- P0/P1/P2/P3 backlog
- 90-day plan
- 12-month plan
- 36-month maturity roadmap

Create:

docs/business/finance/FINANCE_AND_ACCOUNTING_OPERATIONS.md
docs/business/revenue/GTM_AND_REVENUE_OPERATIONS.md
docs/business/trust/SECURITY_PRIVACY_ACCESSIBILITY_COMPLIANCE.md
docs/business/customer/CUSTOMER_SUCCESS_AND_IMPLEMENTATION.md
docs/business/people/PEOPLE_HIRING_AND_CULTURE.md
```

```
docs/business/COMPANY_OPERATING_CADENCE.md
docs/business/COMPANY_KPI_TREE.md
docs/business/BUSINESS_SYSTEMS_OF_RECORD.md
docs/business/BUSINESS_RISK_REGISTER.md
docs/business/BUSINESS_MASTER_BACKLOG.md

For every business process define:

- Trigger
- Owner
- Inputs
- System of record
- Required approval
- Steps
- Output
- KPI
- Failure condition
- Escalation
- Audit trail
- Review cadence
- Documentation
- Next action

Do not treat legal, tax, accounting, insurance, privacy, security, accessibility,
or compliance drafts as final. Clearly flag items needing qualified professional
review.

Return:
- company operating map
- finance operations map
- revenue operations map
- trust/compliance map
- customer-success map
- people/culture map
- exact systems to connect
- dashboard and console requirements
- 90-day execution plan
- top 50 business operations actions in priority order.
```

The business-side finish line is straightforward:

> **Semester becomes credible when it can sell a pilot, implement it safely, support it reliably, prove value, convert it into recurring revenue, protect customer data, retain the customer, and repeat the process without founder heroics.**

Semester Go-to-Market Playbook

Pilot-to-annual institutional sales, financial model assumptions, customer acquisition funnel, compliance/security plan, and customer-success operating workflow.

Purpose. This working playbook helps Semester move from a technically capable education-technology product to a sellable, trustworthy, repeatable business. It is an operating plan—not legal, tax, accounting, insurance, privacy, security, or accessibility advice. Have qualified professionals review customer-facing commitments and binding documents.

Executive direction

Use a two-engine launch model: sell a narrow, measurable institutional workflow through a paid pilot, then use student adoption and verified outcomes as the evidence that converts the pilot to an annual contract and supports expansion.

Do not lead with "replace the university stack." Lead with one urgent workflow, one buyer, a defined cohort, a short timeline, success metrics, and a conversion decision.

1. Launch wedge and offer

Element	Semester pilot definition
Offer	Semester Registration Readiness Pilot
Target buyer	Student Success, Advising, Registrar, or a defined academic division
Problem	Students lack clear next actions before registration; staff manage fragmented guidance, reminders, and support load.
Scope	One department/cohort, one term or 8–12 weeks, limited data/integration scope, named sponsor and champion.
Success	Activation, first meaningful action, workflow completion, staff friction reduction, student confidence/satisfaction, and annual conversion readiness.
Decision	Named economic buyer makes annual conversion decision on an agreed date.

2. Commercial assumptions

Revenue stream	Planning assumption	Guardrail
Student Premium	$8.99/month or $69/year	Validate conversion and support/AI cost before public commitment.
Institution platform	$18 per enrolled student/year; $30,000 annual minimum	Use contract-level volume and scope assumptions.
Implementation	$35,000–$150,000 one-time	Price based on integration, configuration, training, and delivery effort.
AI overage	$30 per 1,000 governed requests	Meter usage; disclose consumption before high-cost actions.
Premium support	15% of annual platform fee; $15,000 minimum	Tie to defined support/response commitments.
Partner/marketplace revenue	12% commission where applicable	Model separately; do not depend on it for core viability.

These are operating assumptions for testing, not promises or universal price terms. Every pilot and annual price should be checked against delivery cost, customer value, procurement feedback, margin, and cash runway.

3. Customer acquisition funnel

Use an account-based B2B funnel for institutional revenue. Student social channels support activation and proof; they are not the primary measurement of enterprise demand.

Stage	Required evidence	Primary KPI
Target accounts	Institution, segment, buyer hypothesis, trigger, priority score	Target-account coverage
Contacted	Personalized outreach, channel, date, contact role	Reply/referral rate
Discovery	Pain, stakeholders, budget, timeline, current workflow	Qualification rate
Demo	Role-specific demo, objections, next action	Demo-to-design rate
Pilot design	Scope, metric baseline, data/integration needs, decision process	Proposal readiness
Security/procurement	Security owner, DPA/HECVAT status, legal/procurement blockers	Review progression
Pilot proposal	Price, annual conversion path, Mutual Action Plan	Proposal-to-pilot close rate
Activated pilot	Admin setup, invitations, adoption baseline	Activation rate
Success achieved	Outcome report and sponsor validation	Pilot KPI attainment
Annual conversion	Annual order form, scope, close date	Pilot-to-annual conversion
Expansion	New cohort/department, reference opportunity	Net revenue retention

Core formulas

• CAC = Sales and marketing spend ÷ new customers acquired.

• CAC payback months = CAC ÷ monthly gross profit from the new customer.

• Stage conversion = accounts entering next stage ÷ accounts entering current stage.

• Pilot conversion = annual contracts converted from pilots ÷ pilots completed.

• LTV sensitivity = annual recurring revenue × gross margin ÷ annual logo churn rate. Use ranges until retention data matures.

• Break-even month = first month in which cumulative gross profit covers cumulative operating expense and required cash obligations.

4. Interactive financial model specification

Build a driver-based 36-month model with monthly columns. Make every forecast assumption editable and visibly distinguish assumed inputs from validated actuals.

Model tab	Required inputs and outputs
Assumptions	Pricing, users per institution, funnel conversion, sales cycle, implementation capacity, cloud/AI cost, payroll, contractor cost, churn, payment terms.
Funnel	Target accounts, contacts, meetings, qualified opportunities, demos, pilot designs, proposals, pilots, annual conversions.

Model tab	Required inputs and outputs
Revenue	Pilot fees, implementation, annual license, student premium, AI overage, support, optional partner revenue.
COGS	Cloud, AI/model usage, storage, support, onboarding/implementation labor, third-party platform costs.
Operating expense	Payroll, contractors, sales/marketing, legal, accounting, insurance, security, software, travel, office/admin.
Cash/runway	Beginning cash, collections timing, burn, scenario runway, financing need, break-even month.
Unit economics	CAC, CAC payback, gross margin, ACV, LTV range, retention and expansion sensitivity.
Scenarios	Conservative, base, ambitious, downside/delayed sales, high-AI-use, pilot-heavy, annual-contract-heavy.

Example monthly driver sequence

Target accounts → outreach → replies → discovery → qualified opportunities → demos → pilot design → proposals → pilots signed → activated pilots → annual conversions. Model collections based on actual payment terms rather than assuming all contract value becomes cash immediately.

5. Institutional pilot sales playbook

Discovery questions

• Walk me through the workflow today. Where do students or staff get stuck?

• What does the problem cost in time, support load, missed actions, or risk?

• Which existing tools and teams are involved?

• Who experiences the pain, who owns budget, and who can block a pilot?

• What is the buying or decision timeline? Is there a triggering event?

• What baseline metric exists today, and what proof would justify annual adoption?

• What security, privacy, accessibility, procurement, and legal review will be required?

Pilot proposal contents

• Executive summary and current-state problem.

• Customer-specific workflow, scope, non-goals, population, timeline, and responsibilities.

• Data/integration assumptions and security/privacy/accessibility boundaries.

• Baseline, success metrics, reporting cadence, midpoint review, and final value review.

• Pilot fee, implementation fee, annual conversion pricing, conversion date, and Mutual Action Plan.

• Known risks, dependencies, support model, and escalation contacts.

Mutual Action Plan minimum fields

Workstream	Customer owner	Semester owner	Completion evidence
Executive sponsorship	Named VP/Dean/CIO	Founder	Goals, decision date, sponsor attendance
Security/privacy	IT/security/privacy reviewer	Security owner	Questionnaire, DPA, data-flow signoff
Contracting	Procurement/legal	Founder/counsel	Pilot agreement/order form

Workstream	Customer owner	Semester owner	Completion evidence
Implementation	Admin/operational champion	Implementation lead	Configured tenant, roles, launch checklist
Training/launch	Champion/comms lead	Customer success	Training complete, communications sent
Outcome review	Economic buyer	Founder/CS	KPI report, annual decision

6. Pilot agreement — business template

This is a commercial outline for counsel to turn into binding language. Do not use it as a signed agreement without higher-ed SaaS counsel review.

Required sections

• Parties and effective date.

• Pilot purpose and defined service/workflow.

• Pilot term, launch milestones, and termination rights.

• Scope, included users/cohort, permitted use, and explicit non-goals.

• Customer responsibilities: champion, access, data, approvals, communications, training attendance.

• Semester responsibilities: configuration, support, reporting, security commitments that are verified, and escalation contacts.

• Fees, invoicing, payment timing, implementation charges, taxes, and expenses.

• Success metrics, baseline, reporting cadence, midpoint review, final review, and non-guarantee language.

• Data protection: roles, data categories, permitted processing, retention, deletion/return, DPA reference, subprocessors.

• Security, incident notification, support, availability, accessibility, and service limits—only as verified and approved.

• Confidentiality, IP ownership, feedback rights, publicity/reference rights, warranties/disclaimers, liability/indemnity, governing law, and order-of-precedence.

• Annual conversion option: price, scope, decision date, and process. Ensure it is optional unless counsel specifies otherwise.

7. Customer success and onboarding

Phase	Target timing	Semester output	Customer outcome
Sales handoff	Immediately after signature	Account plan, scope, risks, success criteria	No surprises between sale and delivery
Welcome + kickoff	Days 0–5	Welcome, agenda, RACI, implementation plan	Shared owners and timeline
Configuration/data	Weeks 1–2	Tenant setup, data/integration validation	Safe, usable configured environment
Training	Weeks 2–3	Admin/champion training, guides, office hours	Operators ready to support users
Launch	Weeks 3–4	Comms kit, onboarding, help center, monitoring	Users invited and activated
Adoption	Weeks 4–8	Usage dashboard, nudges, support triage	Core workflow is used
Midpoint review	Mid-pilot	KPI report, risk plan, executive update	Issues resolved before final review
Results + conversion	Final 2 weeks	Value report, annual proposal, roadmap	Annual decision based on evidence

Customer health score

Signal	Weight	At-risk trigger
Admin engagement	15%	No owner attendance or delayed actions
User activation	20%	Below agreed activation threshold
First meaningful action	15%	Sign-in without value-producing activity
Core workflow completion	20%	Low or declining completion
Weekly active use	10%	Rapid post-launch decline
Support sentiment	5%	Repeated unresolved friction
Executive engagement	5%	Sponsor disengages
Outcome trend	10%	Pilot success metrics stagnate or worsen

8. Cloud security and compliance plan

Operating principle: collect data only for legitimate business need, protect it with administrative, technical, and physical safeguards appropriate to the risk, retain it only as needed, securely dispose of it, and maintain a tested incident response process. [This is a business operating interpretation—not legal advice.]

Control domain	Minimum launch control	Evidence
Governance	Named security owner; risk register; policy review cadence	Security program charter and register
Asset/data inventory	Systems, vendors, data fields, owners, flows, retention	Inventory and data-flow diagrams
Identity/access	MFA; least privilege; joiner/mover/leaver process	Access review evidence
Secrets	Managed secrets; no credentials in Git/client bundles	Secret scan + environment controls
Secure development	PR review; dependency scanning; vulnerability workflow; release checklist	CI reports and remediation log
Data protection	TLS; provider-backed encryption at rest; data minimization	Architecture and configuration evidence
Logging/monitoring	Error monitoring; audit events; alert triage	Dashboard/runbook
Resilience	Backups, restore test, DR runbook	Restore-test record
Incident response	Severity matrix, response team, customer communications, tabletop	IR plan and exercise record
Vendor risk	Subprocessor review and register	Vendor assessments
Higher-ed readiness	HECVAT Lite roadmap and questionnaire library	Completed responses and evidence register
Accessibility	Statement, tested baseline, remediation process	Accessibility evidence and backlog

9. 90-day execution sequence

Window	Primary objective	Deliverables
Days 1–30	Become sellable	Launch wedge, ICP, pricing assumptions, financial model, CRM/account list, sales deck/demo, pilot offer, contract/trust inventory, website conversion paths.

Window	Primary objective	Deliverables
Days 31–60	Become pilot-ready	Pilot agreement through counsel, HECVAT Lite evidence plan, onboarding/training/support workflow, ambassador/student activation kit, first webinars/outbound sequences, pilot proposals.
Days 61–90	Become repeatable	Signed/near-signed pilot, measured outcomes, value report, annual conversion process, case-study process, QBR/renewal cadence, KPI/board rhythm, expansion plan.

10. Owner and meeting cadence

Cadence	Meeting/report	Core measures
Weekly	Revenue review	Accounts, meetings, pipeline, proposal, pilot, close risk
Weekly	Product/engineering review	Delivery, quality, incidents, customer feedback, release risk
Weekly	Customer-success review	Activation, workflow completion, support, health, conversion risk
Weekly	Growth review	Lead conversion, content, channel learning, student activation
Monthly	Finance review	Cash, burn, runway, collections, budget variance, gross margin
Monthly	Security/risk review	Vulnerabilities, vendor changes, evidence gaps, incidents, control owners
Quarterly	Strategy/OKR review	Market learning, product bets, hiring, plan changes, board/investor update

Appendix: Master GTM execution prompt

You are the Chief Revenue Officer, Head of Customer Success, finance lead, marketing lead, security-program coordinator, and operations lead for Semester. Build an execution-ready GTM operating system for an education-technology SaaS company.

Do not create duplicate documents before inventorying existing artifacts. First identify current, stale, duplicate, incomplete, and missing work across sales, marketing, customer success, finance, legal, privacy, security, support, partnerships, and operations.

Build a two-engine model:
1. Institutional B2B revenue through paid, narrow, measurable pilots that convert to annual agreements.
2. Student adoption that creates verifiable activation, workflow-completion, and satisfaction evidence.

Create or update: ICP, positioning, account scoring, 100-account list template, CRM stages, discovery script, demo script, pilot design agenda, pilot proposal, Mutual Action Plan, security/procurement packet, pricing calculator, 36-month financial model, trust-center plan, HECVAT roadmap, customer onboarding, support model, health score, renewal/expansion playbook, student ambassador program, content calendar, KPI tree, weekly operating reviews, and 90-day execution calendar.

Use planning assumptions: Student Premium $8.99/month or $69/year; institution platform $18/enrolled student/year with $30,000 annual minimum; implementation $35,000–$150,000; AI overage $30 per 1,000 governed requests; premium support 15% of annual platform fee with $15,000 minimum; optional partner commission 12%.

For every deliverable, include owner, trigger, inputs, steps, output, system of record, KPI, risks, approval requirements, and acceptance criteria. Clearly label legal, tax, accounting, insurance, privacy, security, accessibility, and contract material that needs qualified professional review. Do not make unverified public claims.

Return: a prioritized P0/P1/P2 backlog, 30/60/90-day plan, weekly scorecard, customer acquisition funnel, financial scenario model specification, trust/compliance evidence register, and first-pilot conversion plan.

Sources used for compliance orientation

FTC guidance highlights data minimization, reasonable safeguards, clear notices, appropriate retention, secure disposal, and incident planning. EDUCAUSE materials describe HECVAT as a higher-education vendor assessment framework. SBA guidance covers business-planning and formation fundamentals. These sources do not replace professional advice.

perplexity

now what else do i need to do create add and genrate on the bussnius and opprations side implentation side sales and marketing side and any other function of a bussnius a company what else to i need to create do build and connect on that deoartment

You need to build **the company operating system around Semester**, not only the product. For an education-technology SaaS business, the remaining work falls into: corporate/legal, finance, security/compliance, product operations, institutional sales, student adoption, marketing, customer success, partnerships, data/analytics, people/HR, and executive governance.

Your current repository already contains unusually extensive product, engineering, audit, security, launch, rollback, retention, and operating documentation. The biggest business-side gap is likely not "more random documents"—it is turning those documents into a **single owned, measurable, recurring operating system** with real pipelines, contracts, dashboards, controls, and customer evidence. [github_mcp_direct]

What to create next

Priority order

Priority	Department	What must exist before serious selling	Outcome
P0	Company/legal	Entity, IP ownership, contracts, privacy terms, data rules	You can legally contract and operate
P0	Security/compliance	Trust Center, security overview, HECVAT evidence, incident process	Institutions can assess you
P0	Finance	Pricing, budget, runway, invoicing, tax/accounting workflow	You know what you sell and whether it is profitable
P0	Sales	ICP, account list, sales process, pilot offer, deck, proposal, CRM	You can create and close pipeline
P0	Marketing	Positioning, site, proof, content, lead capture, lifecycle emails	Buyers can understand and trust Semester
P0	Customer success	Onboarding, support, training, pilot success plan, QBR/renewal process	Customers activate, renew, and refer
P1	Operations	Weekly cadence, metrics, decision log, risk register, vendor management	The company runs predictably

Priority	Department	What must exist before serious selling	Outcome
P1	Product ops	Roadmap governance, feedback system, analytics, release process	Product decisions are evidence-based
P1	Partnerships	Integration strategy, channel partners, student ambassadors	Distribution expands without relying only on outbound
P2	People	Hiring scorecards, handbook, contractors, performance system	You can scale a team safely
P2	Benchmark lab	AI workflow router/benchmark from earlier discussion	Useful internal capability, not a launch blocker

1. Company, legal, and governance

Create these as real, reviewed company artifacts—not just AI drafts:

- Company formation and ownership records
- Founder agreements and equity/vesting documentation
- IP assignment agreements for founders, contractors, employees, and collaborators
- Cap-table system and board/consent records
- Tax registrations and accounting setup
- Business insurance plan:
  - General liability
  - Professional/E&O
  - Cyber liability
  - Directors & officers, when appropriate
- Standard NDA
- Mutual NDA
- Master Services Agreement
- SaaS subscription agreement
- Order form
- Statement of Work template
- Pilot agreement
- Data Processing Addendum
- Privacy Policy
- Terms of Service
- Acceptable Use Policy
- Cookie notice and consent configuration, if applicable
- Subprocessor list
- Service Level Agreement/support policy

- Accessibility statement
- Open-source software policy
- Trademark/name clearance process
- Corporate records folder and contract repository
- Legal-review queue with owner, risk, stage, and outside-counsel escalation criteria

For higher-ed sales, your pilot and contract package should explicitly define: pilot scope, institution roles, data categories, implementation obligations, success criteria, security obligations, accessibility commitment, payment terms, conversion option, renewal terms, termination, and data return/deletion procedure.

Do not publish legal language unreviewed. Use AI to generate structured first drafts and negotiation summaries, then have qualified counsel review terms for the actual entity, customers, data flows, and jurisdiction.

2. Security, privacy, and trust

This is a core sales asset for Semester because institutional buyers will assess vendor risk. HECVAT is specifically designed to help higher-ed institutions evaluate a vendor's cybersecurity, privacy, accessibility, and compliance practices. [1] [2]

Build a Trust Center

Create a public or controlled-access **Trust Center** containing:

- Security overview
- Privacy overview
- Data-processing overview
- Subprocessor list
- Security contact address
- Incident/breach notification commitments
- Availability/status link
- Data deletion and retention summary
- Backup and disaster-recovery summary
- Authentication and access-control summary
- Encryption summary, only for verified controls
- Accessibility statement
- Compliance roadmap and evidence request process
- Download/request flow for security documents under NDA, if needed

Internal security pack

Create and maintain:

- Asset inventory
- Data inventory and data-flow map
- Data classification policy
- Data-retention and deletion schedule
- Access-control policy
- Least-privilege and privileged-access process
- Secure development lifecycle policy
- Vulnerability-management process
- Change-management policy
- Logging and monitoring policy
- Incident-response plan and tabletop exercise
- Business continuity plan
- Disaster-recovery runbook
- Vendor-risk review process
- Secure contractor onboarding/offboarding process
- Security training record
- Penetration-testing roadmap
- Evidence register mapped to controls
- Security questionnaire answer library
- HECVAT Lite first, then a full HECVAT response when target accounts require it

A practical framework is NIST CSF 2.0, which organizes cybersecurity outcomes as **Govern, Identify, Protect, Detect, Respond, and Recover** and explicitly applies to organizations regardless of size or maturity.[3] [4]

Data minimization baseline

For student and institutional data, use the FTC's practical principles:

1. Know what data you collect.
2. Keep only what you need.
3. Protect the data you retain.
4. Dispose of data you no longer need.
5. Maintain an incident-response plan.[5]

3. Finance and business model

You need a real operating model, not only a pricing page.

Create the financial model

Build a 36-month model with:

- Monthly recurring revenue
- Annual contract value
- Pilot revenue
- Implementation/services revenue, if applicable
- Per-seat, per-student, per-department, and enterprise pricing scenarios
- Gross margin
- Cloud/infrastructure costs
- AI/API cost per active user or workflow
- Support and implementation cost
- Sales and marketing spend
- Contractor and employee cost
- Legal, accounting, insurance, and compliance costs
- Burn rate
- Cash runway
- Hiring plan
- Best/base/worst-case scenarios
- Fundraising requirements and timing
- Break-even analysis
- CAC
- CAC payback
- Pipeline conversion assumptions
- Sales cycle assumptions
- Gross retention
- Net revenue retention
- Churn assumptions
- Expansion revenue assumptions

Pricing and packaging

Create:

- A pricing strategy memo
- Packaging architecture:
  - Student/individual, if applicable
  - Department
  - Institution
  - Enterprise/systemwide
  - Pilot
- Feature entitlements per plan
- Usage and overage policy, if relevant
- Implementation fee policy
- Annual-prepay incentive
- Contract length options
- Discount approval policy
- Pricing calculator
- Quote template
- Order form generator
- ROI calculator for institutional buyers
- Procurement justification memo template

For Semester, start with a **narrow, measurable institutional workflow** instead of trying to sell a complete university operating system at once. A buyer should understand the outcome, deployment scope, pilot cost, success metrics, and annual conversion path in one meeting.

4. Institutional sales

Semester needs an institutional **B2B sales engine** plus student adoption. The institutional engine produces revenue; student activity creates proof that the product delivers real outcomes.

Create the sales foundation

- Ideal Customer Profile
- Target vertical and initial beachhead
- Account segmentation:
  - Tier 1: ideal pilots
  - Tier 2: adjacent accounts
  - Tier 3: long-tail/outbound

- Buyer and stakeholder map:
  - CIO/IT
  - Provost/academic leadership
  - Student success
  - Registrar
  - Student affairs
  - Advising
  - Accessibility
  - Security/privacy
  - Procurement
  - Legal
  - Department champion
  - Student champion
- Jobs-to-be-done by buyer
- Problem and ROI narrative by persona
- Competitive battlecards
- Objection-handling library
- Discovery-call script
- Demo script
- Pilot proposal
- Pilot success plan
- Mutual Action Plan
- Security/procurement response kit
- RFP/RFI answer library
- Implementation scope document
- Renewal/expansion playbook
- CRM stages and definitions
- Forecast methodology
- Sales activity standards
- Call-note template
- Account plan template
- Deal-risk review process

Your first sales offer

Define one launch offer in this format:

```
Target customer:
[Specific institution type / department]

Problem:
[One painful workflow with measurable cost, delay, or risk]

Semester workflow:
[What the product does in the first deployment]

Pilot scope:
[Users, departments, integrations, term, implementation timeline]

Success metrics:
[Activation, completed actions, time saved, satisfaction, retention, accuracy]

Commercials:
[Pilot price, implementation fee, annual conversion price]

Conversion decision:
[Named date, decision owner, measurable criteria]
```

Build a 100-account pipeline

Create a CRM target list with:

- Institution
- Enrollment
- Sector/type
- State/region
- Technology stack clues
- Relevant department
- Champion candidate
- Economic buyer
- Security/procurement contact
- Existing tools/alternatives
- Trigger event
- Priority score
- Last activity
- Next step
- Pipeline stage
- Estimated ACV

- Pilot probability
- Close date
- Blocker

Do not measure only demos booked. Measure:

- Qualified accounts
- Discovery conversations
- Security reviews initiated
- Pilot proposals sent
- Pilots signed
- Pilot activation
- Pilot success-metric attainment
- Pilot-to-annual conversion
- Expansion opportunities
- Sales cycle duration
- Win/loss reason

5. Marketing and demand generation

You need one positioning system that powers the website, sales, social, ads, demos, and customer onboarding.

Core brand artifacts

Create:

- Category definition
- Positioning statement
- Ideal customer profile
- Persona messaging
- Messaging hierarchy
- Value proposition matrix
- Brand narrative
- Founder story
- Company boilerplate
- Voice and tone guide
- Claims substantiation register
- Case-study structure
- Proof-point library

- FAQ and objection answers
- Naming/trademark review process
- Design system and visual guidelines

Website requirements

Build or complete:

- Homepage
- Product page
- Specific workflow/use-case pages
- Department/persona pages
- Security/Trust Center
- Pricing or “request pricing” page
- Demo-request page
- Pilot-program page
- Case studies
- Resource center
- Comparison pages
- Careers page, when hiring
- Contact page
- Legal pages
- Status page
- SEO metadata and technical SEO baseline
- Analytics, consent, and conversion tracking
- Lead capture and routing
- Calendar scheduling integration
- CRM integration
- Email nurture trigger after form submission

Content engine

Create:

- 90-day editorial calendar
- SEO topic map
- Keyword-to-page map
- Founder-led content framework
- Customer-proof content framework

- Monthly webinar format
- Email newsletter template
- Case study template
- Research-report template
- Demo-video scripts
- Product explainer video
- Sales clips for social
- Student activation assets
- Press kit
- Media pitch list
- Podcast/outreach list
- Webinar registration and follow-up sequence

Channel strategy

Channel	Primary audience	Main job
Website	All	Explain, qualify, convert
LinkedIn	Institutional buyers	Trust, thought leadership, outbound support
Email	Buyers and leads	Nurture, book meetings, move deals
Webinars	Buyers/champions	Demonstrate expertise and product outcomes
YouTube	Students, buyers, evaluators	Product proof, tutorials, long-form explainers
Instagram/TikTok	Students	Awareness, usefulness, activation
Campus ambassadors	Students	Trust, onboarding, peer adoption
SEO	Future buyers/users	Durable inbound demand
Partnerships	Institutions and channels	Borrowed trust and reach
Paid ads	Retargeting and proven offers	Scale validated conversion—not initial discovery

Do not rely on viral content as the go-to-market model. First prove one institutional workflow works, then turn those measured results into product pages, demos, case studies, webinars, and targeted outbound.

6. Student adoption

Student adoption is a product and operations system, not just social media.

Create:

- Student value proposition
- Student onboarding flow

- “First meaningful action” definition
- Activation checklist
- In-product onboarding
- QR-code and orientation materials
- Faculty/advisor enablement kit
- Student ambassador program
- Ambassador agreement and code of conduct
- Ambassador training
- Ambassador playbook
- Ambassador KPI dashboard
- Campus launch calendar
- Student feedback system
- Referral program, if appropriate
- Student support content
- In-app notification strategy
- Email/SMS policy and consent process
- Re-engagement campaigns
- Student success story template

Track:

- Invited users
- Activated users
- First meaningful action
- Weekly active users
- Workflow completion
- Retention
- Support volume
- Satisfaction
- Referral rate
- Department adoption
- Pilot success outcomes

7. Customer success and support

Create the full post-sale system before scaling sales.

Customer success artifacts

- Customer onboarding plan
- Implementation project plan
- Customer kickoff deck
- Admin setup guide
- Data/import/integration checklist
- Training curriculum
- Training deck
- Office hours plan
- Support center/knowledge base
- In-app help content
- Customer support SLAs
- Ticket categorization
- Escalation policy
- Incident communication templates
- Customer health score
- Adoption dashboard
- QBR template
- Renewal timeline
- Renewal playbook
- Expansion playbook
- Churn-risk playbook
- Customer advisory board charter
- NPS/CSAT program
- Customer feedback-to-roadmap process
- Case-study consent process

Your first pilot should have a written success plan before kickoff:

```
Baseline:
Current process, time, cost, friction, and existing usage

Target:
What “success” looks like by the end of the pilot
```

```
Metrics:
Activation, usage, completion, satisfaction, time saved, errors avoided,
student outcome proxy, staff efficiency, or other buyer-approved metrics

Cadence:
Weekly operating check-in and mid-pilot review

Decision:
Named executive owner, decision date, conversion criteria, next commercial step
```

8. Product operations

Your app repository is strong on audits and technical documentation; now connect product decisions to customer evidence.

Build:

- Product strategy and annual roadmap
- Quarterly OKRs
- Product intake process
- Feature request taxonomy
- Research repository
- Customer feedback tagging
- Opportunity-solution tree
- Prioritization framework
- PRD template
- Technical design document template
- Architecture decision record process
- Experiment template
- Analytics event taxonomy
- Release checklist
- Release notes workflow
- Beta program
- Feature flag governance
- Defect severity policy
- Incident-to-roadmap process
- Product health dashboard
- Data-quality checks
- AI feature evaluation/guardrail process if Semester uses AI

The most important product metrics:

- Time to value
- Activation rate
- First meaningful action
- Weekly/monthly active users
- Workflow completion
- Retention
- Customer health
- Support burden
- Feature adoption
- Pilot-success attainment
- Conversion to annual contract

9. Company operations

Create a lightweight operating cadence that turns all of the above into weekly execution.

Cadence

Cadence	Meeting/report	Purpose
Daily	Founder operating check	Highest-risk decisions, blockers, customer commitments
Weekly	Revenue meeting	Pipeline, deals, next steps, forecast
Weekly	Product/engineering review	Delivery, quality, incidents, customer feedback
Weekly	Customer-success review	Onboarding, health, support, pilot risk
Weekly	Growth review	Funnel, content, campaigns, activation
Monthly	Financial review	Cash, runway, spend, revenue, forecast variance
Monthly	Security/risk review	Vulnerabilities, vendors, incidents, evidence gaps
Monthly	Leadership review	KPIs, decisions, hiring, priorities
Quarterly	OKR and board review	Strategy, resourcing, roadmap, operating plan

Operations artifacts

- Company operating manual
- KPI tree
- Executive dashboard
- Weekly business review format
- Decision log
- Risk register
- RACI matrix

- Operating calendar
- Vendor register
- Procurement process
- Contract repository
- Policy repository
- Corporate records repository
- Budget approval process
- Expense policy
- Invoice and collections workflow
- Renewal calendar
- Data-room index
- Board reporting package
- Investor update template
- Insurance renewal tracker
- Compliance evidence register
- Business continuity plan

10. People and hiring

You may not need a large team now, but you need safe hiring foundations.

Create:

- Org design for the next 12–18 months
- Hiring plan tied to revenue/product milestones
- Role scorecards
- Job descriptions
- Contractor agreements and IP assignment
- Recruiting pipeline
- Interview plans and rubrics
- Reference-check process
- Compensation philosophy
- Equity policy
- Contractor/vendor classification process
- Employee handbook
- Onboarding plan
- Offboarding checklist

- Access-provisioning and deprovisioning process
- Performance-review framework
- Code of conduct
- Whistleblower/complaint escalation process where appropriate
- Security and privacy training

11. Partnerships and ecosystem

For Semester, partners can reduce the cost of trust and distribution.

Build:

- Integration roadmap:
  - SSO/identity
  - LMS
  - SIS/registrar
  - calendar/email
  - analytics
  - data warehouse
  - accessibility tooling
- Partner evaluation rubric
- Integration security review
- Partner agreement template
- Referral partner program
- Campus partner playbook
- Student-success consultant/advisor network
- Implementation partner model
- Channel partner strategy
- University consortium strategy
- Student organization partnership kit
- Ambassador/affiliate policy
- Marketplace listing plan, where applicable
- Integration documentation and developer portal, when ready

12. What to connect

This is the systems layer that prevents every department from operating in isolation.

Business need	Connect to	Purpose
Product users and permissions	Supabase Auth/RLS	Secure customer, institution, and student access
App data and auditability	Supabase/Postgres	Reliable records, tenant scope, event history
Product analytics	Analytics platform + warehouse	Activation, retention, funnel, adoption
CRM	HubSpot, Salesforce, or equivalent	Pipeline, accounts, contacts, forecasting
Website forms	CRM + calendar	Lead routing and meeting conversion
Customer support	Help desk + knowledge base	Ticket tracking and support SLAs
Customer success	CRM/CS platform	Health scores, onboarding, renewals
Billing	Stripe or accounting/invoicing system	Contracts, invoices, payment collection
Accounting	Bookkeeping platform	P&L, cash flow, taxes, reporting
Contract management	Contract repository/CLM	Signed agreements, renewals, obligations
Email	Transactional + lifecycle provider	Product, support, onboarding, sales emails
Status/monitoring	Error tracking + uptime/status platform	Reliability, incident response, transparency
Security	Secret manager, dependency scanning, audit logs	Risk reduction and procurement evidence
Social/content	Scheduling + analytics	Repeatable marketing operations
Data room	Structured Drive/Notion/secure repository	Investor, legal, security, and procurement evidence

Recommended 90-day company build order

Days 1–30: Become sellable

Build:

- One narrow institutional use case
- Clear positioning and ICP
- Website plus demo/pilot page
- Pricing and pilot offer
- CRM and 100-account target list
- Sales deck, demo script, discovery script, proposal
- Legal contract package
- Trust Center minimum viable version
- HECVAT Lite start

- Security overview and questionnaire library
- Financial model and 12-month budget
- Customer onboarding plan
- Product analytics baseline
- Weekly operating dashboard

Days 31–60: Become pilot-ready

Build:

- Pilot agreement and success plan
- Implementation plan
- Customer training materials
- Student adoption kit
- Ambassador pilot
- Support center and escalation process
- CRM lifecycle stages and forecasting
- Email nurture sequence
- First webinar/demo event
- First 25–50 buyer conversations
- First 3–5 pilot proposals
- Product feedback/research loop
- Security evidence pack
- Incident and rollback drills

Days 61–90: Become repeatable

Build:

- Signed pilot or tightly scoped design partner
- Measured pilot outcomes
- Case study or anonymized proof
- Renewal/conversion playbook
- Referral and partner motion
- Improved website from buyer feedback
- Content engine and SEO base
- Board/investor reporting rhythm
- Hiring plan based on actual bottlenecks
- Formalized OKRs and quarterly planning

- Launch/go-no-go review
- Scalable implementation and support playbook

Master business execution command

Paste this into Claude Code, Cursor, or another coding/operations agent **after it has access to your Semester repo and your chosen operating workspace:**

```
You are the Chief of Staff, COO, GTM lead, product operations lead, and security-prog

Semester is an education-technology SaaS company selling to institutions, departments

1. Institutional B2B sales and procurement motion
2. Student adoption and retention motion

Read the existing Semester repository, product documentation, operational documents,

Do not generate duplicate documents that already exist. First create a source-of-tru
- Existing artifact
- Location
- Current owner
- Status: current, stale, duplicate, incomplete, missing, or needs review
- Evidence
- Next action

Then build a unified company operating system across these functions:

A. Corporate, legal, and governance
B. Finance and pricing
C. Security, privacy, compliance, and trust
D. Institutional sales
E. Marketing and demand generation
F. Student adoption
G. Customer success and support
H. Product operations
I. Company operations and executive cadence
J. Partnerships and integrations
K. People and hiring
L. Analytics and data governance

For each function create:
- Objective
- Owner role
- 90-day outcomes
- Required recurring metrics
- Required operating cadence
- Required system of record
- Existing assets
- Missing assets
- Risks
- Dependencies
- P0, P1, P2 backlog
- Acceptance criteria
```

```
Prioritize the following launch-critical deliverables:

1. One clear institutional ICP and one narrow launch workflow
2. Positioning, messaging, website, demo, and pilot offer
3. Pricing, financial model, budget, runway, CAC, and payback model
4. CRM pipeline with 100 named target accounts
5. Sales deck, discovery script, demo script, proposal, pilot agreement, and mutual a
6. Trust Center, security overview, HECVAT Lite evidence plan, security questionnaire
7. Customer onboarding, implementation plan, training, support process, pilot success
8. Student activation kit, ambassador plan, onboarding, first meaningful action, rete
9. Analytics taxonomy and dashboards for revenue, pipeline, activation, retention, ac
10. Weekly business review, revenue review, product review, customer-success review,
11. Partnership and integration strategy
12. Hiring plan tied to actual revenue and product milestones

For every deliverable:
- Reuse and improve existing repository artifacts when possible.
- State whether it is internal, customer-facing, legal-review-required, security-revi
- Add an owner, due sequence, dependency, completion criteria, and review cycle.
- Do not treat AI-generated legal, tax, accounting, privacy, accessibility, or securi
- Do not publish unverified product, security, compliance, accessibility, outcome, or
- Do not build vanity marketing before the ICP, offer, proof, and conversion path are

Create these source-of-truth documents:
- docs/business/COMPANY_OPERATING_SYSTEM.md
- docs/business/90_DAY_COMPANY_BUILD_PLAN.md
- docs/business/GTM_SYSTEM.md
- docs/business/SALES_PLAYBOOK.md
- docs/business/MARKETING_SYSTEM.md
- docs/business/CUSTOMER_SUCCESS_PLAYBOOK.md
- docs/business/FINANCE_AND_PRICING.md
- docs/business/TRUST_AND_COMPLIANCE_ROADMAP.md
- docs/business/STUDENT_ADOPTION_SYSTEM.md
- docs/business/PARTNERSHIPS_AND_INTEGRATIONS.md
- docs/business/PEOPLE_AND_HIRING_PLAN.md
- docs/business/COMPANY_KPI_TREE.md
- docs/business/MASTER_BUSINESS_BACKLOG.md

Create implementation-ready templates where missing:
- ICP template
- Account plan
- Discovery call notes
- Demo script
- Pilot proposal
- Pilot success plan
- Mutual action plan
- Quote and pricing calculator
- Customer onboarding plan
- QBR template
- Security questionnaire response library
- Trust Center content inventory
- Risk register
- Weekly business review
- Investor update
- Board update
- Product feedback record
```

```
- Case-study consent and case-study template
- Student ambassador agreement/training/playbook
- Partner evaluation scorecard
- Vendor risk review
- Job scorecard and interview rubric

At the end, return:
1. Business-function inventory
2. Existing versus missing artifact map
3. P0 launch-critical list
4. 30/60/90-day execution calendar
5. Department ownership matrix
6. KPI tree
7. Required external systems to connect
8. Legal/security/finance items requiring professional review
9. Recommended next 10 actions, in order
10. Files created or updated
```

The single biggest next move

Do these three things first:

1. **Choose one institutional buyer and one workflow** to sell—not "all of higher education."
2. **Build the pilot package**: offer, pricing, deck, demo, security pack, agreement, implementation plan, success metrics, conversion date.
3. **Start 50–100 targeted institutional conversations** while simultaneously building student activation proof.

That sequence turns Semester from a large app and documentation base into a company that can establish trust, close a pilot, prove measurable value, and repeat the motion.

⁂

1. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit
2. https://www.ren-isac.net/hecvat/index.html
3. https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.1300.pdf
4. https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf
5. https://www.ftc.gov/business-guidance/resources/protecting-personal-information-guide-business
6. https://go.wmich.edu/s/article/Higher-Education-Cloud-Vendor-Assessment-Tool-HECVAT
7. https://www.smith-howard.com/how-smbs-can-take-advantage-of-the-nist-cybersecurity-framework-2-0/
8. https://www.purchasing.utah.edu/glossary/hecvat-higher-education-community-vendor-assessment-toolkit/
9. https://www.saltycloud.com/blog/what-is-the-hecvat/
10. https://x.com/NISTcyber/status/1934586756772540924
11. https://www.onestepsecureit.com/blog/navigating-ftc-compliance-a-guide-for-business-owners
12. finance.semester_business_plan
13. projects.semester.gtm_strategy

14. projects.semester.operations_console
15. projects.semester.frontend_company_site
16. https://www.perplexity.ai/search/bde11398-d4d9-44e3-8c2f-f29cfcbf6c8e
17. https://www.perplexity.ai/search/4bbdba62-35e3-4c6e-a35d-1e96479a5e5b
18. https://www.perplexity.ai/search/c0c86bce-5574-48c3-8509-a418276299a8
19. https://www.ftc.gov/business-guidance/privacy-security
20. https://www.nist.gov/itl/smallbusinesscyber/nist-cybersecurity-framework-0
21. https://cloud.google.com/security/compliance/hecvat
22. https://www.zoom.com/en/trust/legal-compliance/hecvat/

now expand on how semester aims to be the chatgpt claude perplexity top hat bright space canvas SIS registar universy systems services google apple microsoft and central education OS and platform want to become like google and micofot for education expand in how to become that entire point and thiesis is that it replaces everything from its inception

Semester's thesis can be much bigger than "an AI study tool" or "a better university portal." The full ambition is to become the **central operating system for education**: the unified platform through which students live their educational journey and institutions run the systems, workflows, intelligence, services, and decisions that support it.

The strategic doctrine is:

> **Connect first. Replace by domain. Operate as one system.**

Semester can begin by interoperating with existing systems where institutions require continuity, then progressively become the system of record and operating layer for each domain as capability, trust, migration readiness, and institutional authorization mature. Education interoperability standards such as LTI, OneRoster, and Edu-API exist specifically to connect learning tools, roster/course data, and administrative/teaching systems in a more secure and standardized way.[1] [2] [3]

The complete Semester thesis

One platform, many roles

Semester is intended to be all of the following as **one coherent education platform**, not a loose bundle of apps:

Role	What Semester becomes
Student OS	The daily home for identity, schedule, planning, tasks, learning, decisions, progress, support, and opportunity
Productivity suite	Notes, tasks, calendar, planning, files, collaboration, messaging, focus, and workflow automation
Learning platform	Courses, content, assignments, feedback, study tools, assessment support, grade visibility, tutoring, and AI-guided learning

Role	What Semester becomes
Institutional OS	Tenant configuration, policy, identity, roles, workflows, academic operations, student services, analytics, compliance, and rollout controls
AI platform	Governed assistants, search, research, tutoring, planning, advising, agent workflows, evaluation, policy controls, and human oversight
Campus super-app	Housing, dining, events, organizations, student accounts, campus services, support, career, safety, and community
Academic record layer	Controlled records, registrar workflows, enrollment, registration, grades, progression, degree planning, transcripts/records integration or native operation
Marketplace/platform	Approved integrations, institutional services, learning resources, partners, career opportunities, and commerce where appropriate
Family/guardian platform	Student-consented, minimum-necessary, time-bounded views and support pathways
Institutional intelligence layer	Aggregate outcomes, demand signals, service bottlenecks, retention risk, capacity planning, and decision support
Company platform	Sales, pilots, customer success, billing, trust, compliance, support, release management, and business operations

The product should feel like **one education graph and one daily workspace:**

```
Identity
  ↓
Path
  ↓
Plan
  ↓
Courses
  ↓
Actions
  ↓
Support
  ↓
Progress
  ↓
Career
  ↓
Lifelong relationship
```

A student should not experience Semester as 20 portals. They should experience:

> "This is where I understand where I am, what matters next, what I can do, who can help, and how I move forward."

What Semester replaces

AI and intelligence layer

Semester aims to replace the fragmented educational use of ChatGPT, Claude, Perplexity, generic search, scattered tutoring tools, and unmanaged AI experiments—not necessarily by forbidding external models, but by making them operate through a unified governed education layer.

Fragmented tool pattern	Semester replacement
ChatGPT used without course context	Course-aware AI assistant with policy, source, and instructor boundaries
Claude used as isolated writing/reasoning workspace	Semester study, planning, drafting, research, and project workspace
Perplexity used for broad research	Source-aware research workflow with citations, approved sources, and institution policy
Separate AI tutor apps	Student-path-aware tutoring and study support
Separate advising chatbot	Explainable, policy-bounded student guidance with human handoff
Generic AI agents	Permissioned education agents with audit trails, approval gates, and data classification
Uncontrolled AI data sharing	Tenant-level provider policy, data minimization, retention rules, and spend controls
AI use without academic integrity context	Course-level AI rules, disclosure guidance, permitted-use boundaries, and educator controls

The AI architecture should be **model-flexible but Semester-governed**:

```
Student / Staff / Institution
        ↓
Semester AI Gateway
        ↓
Policy check
- Role and tenant permissions
- Data classification
- Course/institution AI rules
- Consent
- Budget and rate limit
- Approved source policy
- Integrity requirement
        ↓
Model and tool routing
- Internal capability
- Approved external model
- Search/research provider
- Institutional knowledge source
- Human escalation
        ↓
Evidence, citations, logs, and outcome
```

NIST's AI Risk Management Framework is relevant here because it is designed to help organizations incorporate trustworthiness into the design, development, deployment, use, and evaluation of AI systems. [4] [5]

Learning management and courseware

Semester ultimately aims to replace or absorb the role played by Canvas, Brightspace, Blackboard, Top Hat, courseware tools, separate study apps, disconnected assignment systems, and fragmented class communication.

```
Course Studio
├── Course home
├── Learning objectives
├── Instructor guidance
├── Course AI rules
├── Resources and approved sources
├── Assignments
├── Study packs
├── Notes and annotations
├── Discussions and collaboration
├── Attendance/participation, where authorized
├── Assessment workflows
├── Feedback
├── Gradebook
├── Regrade process
├── Integrity and provenance controls
├── Learning analytics
└── LTI / external-tool transition support
```

The native Course Studio must not merely mimic an LMS page. It should connect the course to the student's actual plan, schedule, workload, goals, support needs, career interests, and institutional policies.

SIS, ERP, registrar, and academic records

Semester's long-term goal is to replace fragmented SIS, registrar, academic planning, records, billing, degree audit, and student account workflows with one canonical operating layer.

```
Academic and registrar core
├── Identity and affiliation
├── Programs and pathways
├── Terms and calendars
├── Course catalog
├── Sections and schedules
├── Enrollment
├── Waitlists
├── Holds
├── Prerequisites
├── Overrides
├── Registration
├── Degree planning
├── Transfer evaluation
```

```
├── Articulation rules
├── Academic records
├── Grade ledger
├── Regrades
├── Graduation scenarios
├── Official handoffs and exports
└── Audit and compliance history
```

The migration path matters:

```
Phase 1: Connect
Existing SIS/ERP remains authoritative.
Semester reads approved, minimum-required data.

Phase 2: Co-pilot
Semester becomes the student and staff action layer.
Institution validates workflow outcomes and data quality.

Phase 3: Domain replacement
Semester natively operates selected domains:
- planning
- advising workflow
- registration readiness
- student action center
- course experience
- community
- support
- career
- communications

Phase 4: Core-system replacement
Semester becomes the authoritative system of record for approved domains,
with migrations, dual-run validation, audit, reconciliation, and rollback.

Phase 5: Education operating system
Semester becomes the primary system through which university operations,
learning, student life, administration, intelligence, and services run.
```

The correct end state is not “Semester always sits beside the SIS.” The end state is:

> Semester becomes the authoritative platform where the institution chooses it to be authoritative, while preserving data portability, official auditability, interoperability, and controlled transition paths.

Productivity and collaboration

Semester should replace the education-specific use of Google Workspace, Microsoft 365, Apple productivity workflows, Notion, task tools, calendar tools, note apps, group-chat tools, and scattered student productivity systems.

```
Semester Workspace
├── Personal dashboard
```

```
├── Academic calendar
├── Tasks and action center
├── Notes and knowledge capture
├── Documents and files
├── Planner
├── Projects
├── Group work
├── Messaging
├── Reminders
├── Study sessions
├── Focus tools
├── Research workspace
├── AI workspace
├── Portfolio
├── Career workspace
└── Cross-device sync
```

The difference is contextual integration:

```
Google/Microsoft/Apple-style productivity:
General tools → user adapts them to education.

Semester productivity:
Educational context → tools adapt to the student's academic reality.
```

A task is not merely "write paper." It can be connected to:

- A course
- A learning objective
- A due date
- An instructor's policy
- Study resources
- A support office
- A degree requirement
- A career competency
- A student's workload and wellbeing constraints

Search and knowledge

Semester should replace fragmented web search, campus search, course search, document search, policy search, support search, and institutional knowledge silos.

```
Semester Search
├── Personal workspace search
├── Course content search
├── Institution policy search
├── Student-service search
├── Campus-resource search
```

```
├── Career and opportunity search
├── Library/research search
├── Approved-source search
├── AI-assisted answer generation
├── Citation and provenance
├── Permission-aware results
└── Human handoff when confidence is insufficient
```

The search system must distinguish:

Source type	Semester behavior
Official institution record	Clearly labeled as official and source-authoritative
Instructor/course material	Scoped to course and term
Student-owned content	Private by default
Approved learning source	Labeled with owner/version
Public web/research	Cited, time-aware, policy-controlled
AI-generated guidance	Explained, attributable, and non-authoritative unless verified
Community content	Moderated and clearly distinguished from official guidance

Semester as Google and Microsoft for education

The comparison is useful if it is understood correctly.

Semester does **not** become "Google with a campus theme" or "Microsoft 365 for students." It becomes the infrastructure layer that combines what those companies do best—identity, productivity, search, collaboration, developer ecosystems, cloud operations, and AI—with education-specific authority, workflows, protections, and outcomes.

Platform model	What it provides	Semester equivalent
Google	Search, identity, workspace, cloud, AI, ecosystem	Education search, education identity, student workspace, institutional cloud/control plane, governed AI, education ecosystem
Microsoft	Productivity, identity, collaboration, enterprise admin, developer platform	Academic productivity, student/staff identity, course/community collaboration, institutional administration, education developer platform
Apple	Integrated hardware/software experience, privacy, premium UX	Calm integrated cross-device education experience, student agency, privacy-centered product design
ChatGPT/Claude/Perplexity	General-purpose reasoning, writing, research, conversation	Governed educational intelligence connected to courses, policies, sources, plans, and human support
Canvas/Brightspace/Blackboard/Top Hat	Course delivery and learning workflows	A connected course experience integrated into every other education workflow

Platform model	What it provides	Semester equivalent
SIS/ERP/Registrar systems	Institutional records and transactions	A unified education record, action, policy, and workflow engine
Student success point tools	Advising, nudges, alerts, retention	One coherent student-path and intervention system
Campus apps	Dining, housing, events, transit, organizations	Unified campus life and services layer
CRM/operations tooling	Sales, support, renewal, billing	Internal company command center for running Semester itself

The education graph

The deepest defensibility is not any one feature. It is the **education graph.**

```
Person
├── Identity
├── Institution
├── Program
├── Courses
├── Terms
├── Requirements
├── Skills
├── Tasks
├── Deadlines
├── Resources
├── Instructors
├── Advisors
├── Support offices
├── Communities
├── Financial obligations
├── Housing/dining/campus services
├── Career opportunities
├── Family/guardian permissions
├── AI interactions
├── Evidence/provenance
└── Outcomes
```

Every interaction strengthens the graph:

```
Student plans a course
→ plan updates workload
→ workload changes task suggestions
→ task connects to course objectives
→ course connects to study resources
→ missed task creates support signal
→ support signal creates explainable intervention option
→ intervention outcome informs institutional insight
→ student still controls private content and sharing boundaries
```

This is why Semester can become difficult to replace: it is not merely a collection of features. It becomes the connected operational memory of the educational journey.

Non-negotiable operating principles

To replace everything responsibly, Semester must be stronger than the systems it replaces in these areas:

1. Authority

```
Official records must have a defined authoritative source.
No AI output becomes an official record by accident.
No external source is silently treated as official.
Every high-impact institutional action is attributable and auditable.
```

2. Tenant isolation

```
Institution A cannot access Institution B.
One school's policy cannot accidentally apply to another.
Student data is scoped to student, tenant, role, grant, and purpose.
Cross-tenant analytics is aggregated, approved, and privacy-governed.
```

3. Student agency

```
Student-owned workspace data is private by default.
Family/advisor/support sharing is explicit, scoped, time-bounded, revocable,
and logged.
Students see what is shared, with whom, and why.
```

4. Interoperability

```
No lock-in through trapped data.
Import/export exists.
Institutional systems can integrate through governed APIs.
Migration paths are reversible until formal cutover.
Standards are adopted where practical:
- LTI
- OneRoster
- Edu-API
- SSO
- SCIM
- secure APIs
```

LTI is designed to connect learning tools securely with institutional learning environments, including user, enrollment, and role information. OneRoster supports secure exchange of users, courses, enrollments, learning resources, and grades across education systems. [2] [6] [1]

5. Explainability

```
Semester can explain:
- why a recommendation appears
- what data was used
- what source is authoritative
- what policy constrained the result
- what the user can change
- when human review is needed
```

6. Governance

```
Every major configuration has:
- owner
- policy
- approval path
- evidence
- effective date
- audit history
- rollback plan
```

7. Safety and resilience

```
Core education workflows must survive:
- vendor outage
- failed sync
- bad import
- release regression
- mistaken configuration
- compromised account
- privacy request
- data retention deadline
- institution offboarding
```

The product architecture

```
Semester Education OS
│
├── Experience Layer
│   ├── Student OS
│   ├── Faculty/Course Studio
│   ├── Advisor and Student Success Workspace
│   ├── Registrar/Academic Operations
│   ├── Campus Services
│   ├── Family/Guardian Portal
│   ├── Employer/Partner Portal
│   └── Semester Company Operations Console
│
├── Intelligence Layer
│   ├── AI Gateway
```

```
│   ├── Policy engine
│   ├── Search and knowledge graph
│   ├── Recommendation engine
│   ├── Workflow automation
│   ├── Analytics and outcomes
│   └── Evidence/provenance engine
│
├── Domain Layer
│   ├── Identity and roles
│   ├── Academic records
│   ├── Course and learning
│   ├── Registration
│   ├── Degree planning
│   ├── Student accounts
│   ├── Campus services
│   ├── Support
│   ├── Community
│   ├── Career
│   ├── Family/guardian
│   ├── Billing and commercial
│   └── Governance/compliance
│
├── Control Layer
│   ├── Tenant policy
│   ├── Approval system
│   ├── Audit trail
│   ├── Break-glass access
│   ├── Feature flags
│   ├── Rollout controls
│   ├── SLOs/incident response
│   ├── Privacy/retention
│   └── Trust/compliance evidence
│
└── Integration Layer
    ├── SIS/ERP
    ├── LMS
    ├── LTI tools
    ├── SSO/SCIM
    ├── Productivity suites
    ├── Library/research systems
    ├── Payment providers
    ├── Identity providers
    ├── Campus services
    ├── External AI providers
    └── Partner ecosystem
```

Product doctrine

Use this as the central internal statement:

> **Everything belongs in Semester—but nothing should feel bolted on.**

Every feature must answer:

1. Which role uses it?

2. Which domain owns the data?
3. What is the source of authority?
4. Which tenant policy governs it?
5. Which permissions are required?
6. What is the student/teacher/staff value?
7. How does it connect to the education graph?
8. How is it audited, supported, tested, and rolled back?
9. Does it replace, connect to, or hand off to an external system?
10. What is the migration path from external dependency to native Semester operation?

Migration model: connect, replace, operate

Phase 1: Connect

Semester connects to existing fragmented systems.

```
Existing tools:
SIS
LMS
Email
Calendar
Cloud storage
Library
Payment providers
Campus card
Housing
Dining
Career tools
External AI
```

Semester provides:

- One identity-aware workspace
- One action center
- One planning layer
- One search layer
- One student journey
- One support route
- One governed AI interface
- One institutional control plane

Phase 2: Replace visible workflows

Semester replaces the fragmented **experience** first:

- Student portal
- Planner
- Advising action center
- Registration readiness
- Course workspace
- Campus event/service navigation
- Support intake
- Community
- Career action center
- AI assistant
- Institutional workflow approvals

Phase 3: Replace operational domains

Semester becomes native system of record for selected domains:

- Productivity workspace
- Course Studio
- Student tasks/actions
- Student support workflows
- Institutional communications
- Community
- Campus services
- Career opportunity workflows
- Student success operations
- Tenant policy
- Commercial/customer lifecycle

Phase 4: Replace core institutional systems

Only when governance, migrations, auditability, legal review, operational readiness, and institutional approval exist:

- Registration
- Academic records
- Gradebook

- Student accounts
- Payment plans
- Degree progression
- Institutional analytics
- Core identity/provisioning
- Multi-campus governance
- Enterprise education operations

How to communicate this externally

Do not say:

> “Semester replaces Google, Microsoft, ChatGPT, Canvas, Brightspace, the SIS, the registrar, and every university system tomorrow.”

Say:

> “Semester is building the unified operating system for education: one connected platform for the student journey, learning, academic operations, campus services, institutional intelligence, and governed AI. It can connect to existing systems during transition and become the authoritative operating layer for domains institutions choose to modernize.”

For a pilot:

> “Semester begins with one measurable workflow—such as registration readiness—then proves student and institutional value before expanding across the education operating system.”

That protects the full thesis without making an unbuyable first sales claim.

Master product-thesis prompt

```
You are the Chief Product Officer, Chief Technology Officer, Chief Learning
Officer, Chief Information Security Officer, Chief Operating Officer, and
enterprise education architect for Semester.

Semester’s thesis is not to become another point solution, AI wrapper, LMS
plugin, or student planner.

Semester is building the unified operating system for education.

Its long-term mission is to become the central platform through which students,
institutions, faculty, staff, families, partners, and employers experience,
operate, govern, and improve education.

Semester must unify and progressively replace the fragmented functions now
spread across:
```

```
- ChatGPT
- Claude
- Perplexity
- generic AI assistants
- Google Workspace
- Microsoft 365
- Apple productivity workflows
- Canvas
- Brightspace
- Blackboard
- Top Hat
- courseware tools
- SIS
- ERP
- Registrar systems
- academic records
- degree audit
- registration systems
- student accounts
- payment plans
- advising tools
- student success tools
- tutoring tools
- campus apps
- housing
- dining
- events
- student organizations
- career platforms
- support portals
- CRM and operational tools
- family/guardian portals
- fragmented university service systems

The strategy is:

CONNECT FIRST.
REPLACE BY DOMAIN.
OPERATE AS ONE SYSTEM.

Do not narrow the long-term product thesis. The platform must be designed from
its inception as:

- Student OS
- Productivity suite
- Campus super-app
- Course and learning platform
- Institutional operating system
- AI and intelligence platform
- Academic/registrar platform
- Marketplace and partner ecosystem
- Family/guardian platform
- Career and lifelong learner platform
- SaaS company operating platform

However, do not implement it as disconnected feature modules. Build one coherent
```

```
education graph and one role-aware operating system.

For every domain, define:

1. User roles
2. Jobs to be done
3. System of record
4. Source authority
5. Tenant boundaries
6. Permission model
7. Data classification
8. AI policy
9. Integrations required during transition
10. Native replacement capability
11. Migration path
12. Audit requirements
13. Rollback path
14. Support model
15. Compliance requirements
16. Success metrics
17. Commercial model
18. Launch readiness criteria

Build or update:

docs/product/SEMESTER_UNIFIED_EDUCATION_OS_THESIS.md
docs/product/EDUCATION_GRAPH_ARCHITECTURE.md
docs/product/DOMAIN_REPLACEMENT_ROADMAP.md
docs/product/CONNECT_REPLACE_OPERATE_STRATEGY.md
docs/product/AI_GOVERNANCE_AND_MODEL_ROUTING.md
docs/product/LEARNING_AND_COURSE_STUDIO.md
docs/product/ACADEMIC_RECORDS_AND_REGISTRAR.md
docs/product/STUDENT_PRODUCTIVITY_AND_WORKSPACE.md
docs/product/CAMPUS_SERVICES_PLATFORM.md
docs/product/FAMILY_GUARDIAN_MODEL.md
docs/product/CAREER_AND_LIFELONG_LEARNING.md
docs/product/INSTITUTIONAL_CONTROL_PLANE.md
docs/product/INTEROPERABILITY_AND_MIGRATION_STRATEGY.md
docs/product/EDUCATION_OS_ROLE_MATRIX.md
docs/product/EDUCATION_OS_DATA_AUTHORITY_MATRIX.md
docs/product/EDUCATION_OS_RELEASE_GATES.md

For every replacement claim:
- distinguish current implementation, planned capability, integration, pilot,
  native system of record, and future roadmap
- do not imply a native capability exists if it does not
- do not make a compliance, security, accessibility, or institutional authority
  claim without evidence
- do not replace official records, grades, financial accounts, identity,
  registration, or legal processes without formal governance, migration,
  authorization, audit, rollback, and institutional approval

The final product must make users feel:
“This is not another portal. This is where my educational life works.”
```

The central one-sentence thesis

> **Semester is the unified, AI-native operating system for education: one trusted platform that connects and progressively replaces the fragmented tools, portals, learning systems, institutional records, campus services, productivity suites, AI assistants, and operational workflows that currently make education difficult to navigate and operate.**

⁂

1. https://www.1edtech.org/standards/lti
2. https://www.1edtech.org/standards/oneroster
3. https://www.1edtech.org/standards/edu-api
4. https://www.nist.gov/itl/ai-risk-management-framework
5. https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
6. https://www.1edtech.org/blog/rostering-resources-and-gradebook-standards
7. https://www.1edtech.org/specifications
8. https://er.educause.edu/articles/2026/2/harnessing-data-taming-digital-sprawl-and-enabling-experiential-learning-for-student-success
9. https://www.imsglobal.org/about/k12/oneroster-and-lti-advantage
10. https://www.imsglobal.org/specs/ltiv1p1
11. https://www.imsglobal.org/learning-tools-interoperability-resources
12. https://www.imsglobal.org/oneroster-conformance
13. https://www.educause.edu/content/2026/the-integrated-institution-a-guide-to-interoperability-for-student-success
14. https://www.imsglobal.org/pressreleases/pr120501.html
15. https://www.1edtech.org/1edtech-article/1edtech-demonstrates-global-power-of-interoperability-standards/411320
16. projects.semester.product_vision
17. https://www.perplexity.ai/search/e9bf631e-d220-4caa-af31-a0ae2586458f
18. https://www.perplexity.ai/search/2726514c-0e6d-40f9-a547-3dc49ae6208d
19. projects.semester.architecture_scope
20. goals.semester.market_positioning
21. https://www.perplexity.ai/search/82cf1129-7986-4186-bee5-3c81329acd24
22. https://www.perplexity.ai/search/70a6c7ac-df0f-46ab-9d6e-dfc502a148a0
23. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit
24. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/hecvat-faqs-for-higher-education
25. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/how-to-use-the-higher-education-community-vendor-assessment-toolkit
26. https://www.educause.edu/higher-education-community-vendor-assessment-toolkit/hecvat-faqs-for-corporations
27. https://er.educause.edu/articles/2025/2/hecvat-4-better-than-ever
28. https://www.ren-isac.net/services/hecvat.html

29. https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf

30. https://www.nist.gov/video/introduction-nist-ai-risk-management-framework-ai-rmf-10-explainer-video

31. https://library.educause.edu/~/media/files/library/2014/9/erb1411-pdf.pdf

perplexity

expand on eveything you can possiably thing of to build run oprate mange update opprate Semmester SaaS whole point to be the google apple micosoft chat gpt claud perplexity quizlit brightspace canvas regisar sis market place social connection universty OS and Company of Education provide and rendr eveything needed for semester and further enhance expand and imporve Semester is the education operating system that replaces the fragmented infrastructure of learning, academic life, student support, campus services, institutional operations, productivity, and AI with one connected, governed, human-centered platform Build everything natively. Integrate responsibly. Replace only with proof. Operate every domain as one system.

Semester should be designed and operated as a **complete education infrastructure company**, not simply an app, LMS, AI feature, student portal, or SaaS dashboard. Its total scope is the connected operating system for learning, academic life, institutional administration, campus services, productivity, AI, data, trust, commerce, and lifelong outcomes.

The implementation doctrine remains:

> **Build every core capability natively. Integrate responsibly during transition. Replace each domain only with proof. Operate every domain as one coherent system.**

The prior architecture already establishes the main domains—student OS, learning/Course Studio, AI gateway, student success, registrar, finance, campus services, career, family/guardian, institutional control plane, and ecosystem. This response expands the missing company, platform, reliability, governance, research, operations, data, commercial, people, and developer layers required to make that thesis real. [1]

Semester’s complete operating model

The 12 systems Semester must become

```
1. Education Experience System
2. Learning and Assessment System
3. Academic Records and Registrar System
4. Student Success and Support System
5. Productivity, Knowledge, and Collaboration System
6. Governed AI and Intelligence System
7. Campus Life and Commerce System
8. Career, Credential, and Lifelong Learning System
9. Identity, Trust, Privacy, and Governance System
10. Institutional Operations and Intelligence System
11. Platform, Developer, and Partner Ecosystem
12. Semester Company Operating System
```

Every system must connect to the same:

```
Identity
→ Tenant
→ Role
→ Capability
→ Policy
→ Data classification
→ Consent
→ Source authority
→ Workflow
→ Audit
→ Evidence
→ Outcome
```

That shared foundation is what makes Semester an operating system rather than a bundle of products.

1. Education experience system

The education experience system is the interface layer for every major role.

Role	Primary Semester experience	Daily value
Student	Student OS	Understand current status, make decisions, complete actions, learn, get support
Faculty	Course Studio	Teach, publish guidance, manage learning, assess, communicate, govern AI use
Advisor	Student Success workspace	Understand student path, coordinate support, manage actions and referrals

Role	Primary Semester experience	Daily value
Registrar	Academic Operations	Manage terms, catalog, enrollment, holds, rules, records, and graduation workflows
Institution leader	Institutional intelligence	See aggregate outcomes, risk, adoption, service bottlenecks, and operational health
Family/guardian	Consent-based support portal	Receive only student-authorized, time-bounded information and support paths
Employer	Talent and opportunity portal	Discover verified talent evidence and connect to candidates appropriately
Partner	Integration/marketplace portal	Build governed integrations, extensions, resources, and services
Semester operator	Operations Command Center	Run tenants, pilots, support, security, billing, releases, compliance, and company operations

Experience requirements

Every role-specific surface needs:

- Clear primary job-to-be-done.
- Role-aware navigation.
- Correct tenant scope.
- Minimum necessary information.
- Source authority labels.
- Data freshness state.
- Accessibility support.
- Human escalation path.
- Auditability for sensitive actions.
- Mobile and desktop behavior.
- Clear distinction between:
  - official record
  - guidance
  - recommendation
  - student-owned content
  - institutional content
  - external source
  - AI-generated content
  - community-generated content

2. Learning and assessment system

Semester must become a full learning environment, not merely a course-content viewer.

```
Learning OS
├── Course Studio
├── Curriculum map
├── Learning objectives
├── Syllabus and policies
├── Course schedule
├── Assignments
├── Resources
├── Readings and annotations
├── Notes
├── Practice
├── Flashcards
├── Study packs
├── Peer learning
├── Tutoring
├── Discussions
├── Office hours
├── Assessment
├── Feedback
├── Gradebook
├── Regrade process
├── Academic integrity
├── Accessibility accommodations
├── Course AI rules
├── Learning evidence
├── Competency map
└── Course improvement analytics
```

Further capabilities to build

Learning design

- Course templates.
- Curriculum-to-course mapping.
- Program learning outcome mapping.
- Objective-to-assignment alignment.
- Objective-to-assessment alignment.
- Rubric builder.
- Competency framework.
- Mastery progression.
- Prerequisite knowledge maps.
- Adaptive study recommendations.
- Workload estimation.

- Assessment calendar load balancing.
- Course versioning by term.
- Course archive/export.
- Course accessibility review workflow.
- Course policy publication workflow.
- Content approval and licensing records.

Assessment and integrity

- Native assessments.
- Question banks.
- Rubrics.
- Blind grading where configured.
- Moderation workflow.
- Regrade requests.
- Grade release controls.
- Assessment accommodations.
- Assignment submission provenance.
- Academic-integrity guidance.
- Course-specific AI allowed/disallowed uses.
- Citation assistance.
- Draft disclosure tools.
- Plagiarism/integrity integration where approved.
- Human review queue.
- Appeal workflow.
- Immutable grade version history.
- Grade passback controls during transition.

Learning analytics

Do not reduce learning to engagement surveillance. Use explainable, purpose-limited signals:

```
Learning signals
├── Submitted work
├── Objective completion
├── Practice behavior
├── Study-plan progress
├── Optional self-checks
├── Instructor feedback cycles
├── Resource usefulness
```

```
├── Support requests
└── Student-reported confidence
```

Avoid hidden scoring of students based on unrelated behavioral data. Students and institutions should understand what signals exist, why they are used, and how they affect recommendations.

3. Academic records and registrar system

Semester must eventually operate a complete academic transaction system.

```
Academic Core
├── Institutions
├── Campuses
├── Schools
├── Departments
├── Programs
├── Credentials
├── Terms
├── Catalog
├── Courses
├── Sections
├── Meetings
├── Instructors
├── Rooms
├── Enrollment
├── Waitlists
├── Holds
├── Prerequisites
├── Overrides
├── Registration windows
├── Time tickets
├── Add/drop
├── Withdrawal
├── Grades
├── Academic record
├── Transfer credit
├── Articulation agreements
├── Degree audit
├── Graduation scenarios
├── Conferral
├── Transcript/export
└── Institutional reporting
```

Academic-system design requirements

Every academic operation requires:

Requirement	Why
Idempotency	Prevent duplicate registration, grade, payment, or record changes
Source authority	Identify whether Semester or an external system is official

Requirement	Why
Effective dates	Rules and records change over terms/time
Immutable history	Preserve grade, enrollment, and record changes
Two-person approvals	Required for high-risk corrections or system migration
Reconciliation	Compare old and new system outcomes
Capacity correctness	Prevent over-enrollment and waitlist corruption
Policy versioning	Academic rules must be traceable to version/effective date
Audit trail	Explain who changed what, when, and why
Rollback	Restore from failed migration/configuration
Exportability	Ensure institution/data portability
Privacy controls	Limit access to educational records

Native academic rules engine

Build a deterministic rules engine for:

```
Eligibility
Prerequisites
Co-requisites
Credit limits
Program requirements
Residency requirements
Transfer equivalencies
Academic standing
Holds
Time tickets
Waitlist promotion
Graduation requirements
Policy exceptions
```

AI can explain an academic rule, but it must not be the final rule engine.

```
Deterministic policy engine
→ computes official eligible/ineligible state

AI copilot
→ explains the decision in plain language
→ identifies next steps
→ links to official rule/policy
→ routes exceptions to humans
```

4. Student success and support system

Semester should make support proactive without becoming invasive.

```
Student Success OS
├── Action Center
├── Advising
├── Case management
├── Appointment scheduling
├── Check-ins
├── Success plans
├── Academic planning
├── Registration readiness
├── Referral workflows
├── Financial-support handoffs
├── Accessibility-service handoffs
├── Tutoring
├── Peer mentoring
├── Emergency/safety handoffs
├── Office action feeds
├── Support ticketing
├── Knowledge base
├── Communications
├── Outreach
├── Intervention review
├── Outcome tracking
└── Aggregate institutional insights
```

Support model

```
Student signal or request
  ↓
Student sees available self-service information
  ↓
Student can take action or request help
  ↓
Correct office/team receives structured request
  ↓
Owner accepts, routes, resolves, or escalates
  ↓
Student receives status and next steps
  ↓
Outcome is recorded without excessive sensitive detail
  ↓
Aggregate data improves service design
```

Support principles

- Student asks for help without navigating institutional org charts.
- Support requests retain context only when necessary.
- No broad administrator visibility into private student material.
- Referrals require policy and purpose.
- Sensitive support categories require stronger controls.
- Emergency/safety actions follow institution-specific official protocols.
- Students can see the status of their own requests.
- Support quality is measurable:
  - response time
  - resolution time
  - satisfaction
  - repeat issue rate
  - escalation rate
  - outcome completion

5. Productivity, knowledge, and collaboration system

Semester's productivity layer replaces generic tools through education context—not through superficial clones.

```
Semester Workspace
├── Personal knowledge base
├── Notes
├── Tasks
├── Calendar
├── Documents
├── Files
├── Projects
├── Groups
├── Shared workspaces
├── Study sessions
├── Focus timer
├── Research workspace
├── Writing workspace
├── Course notebooks
├── Meeting notes
├── Decision records
├── Version history
├── Search
├── Automations
├── Templates
└── Cross-device sync
```

Capabilities that make it education-native

Generic productivity tool	Semester-native upgrade
To-do item	Connected to course, deadline, learning objective, support office, academic requirement, and workload
Calendar event	Connected to course, term, assessment, registration window, appointment, campus event, and action
Note	Connected to course, reading, objective, assignment, AI citation, and study resource
File	Connected to ownership, course, privacy policy, retention, sharing, and academic integrity
Project	Connected to group roles, course rubric, milestones, feedback, and assessment requirements
Reminder	Connected to urgency, dependency, institutional deadline, and user preference
Search	Permission-aware across personal, course, institutional, and approved public sources
Collaboration	Course/group/workspace-specific with appropriate membership and visibility

6. AI and intelligence system

Semester must build a complete **Education Intelligence Platform**, not a generic chatbot panel.

```
Education Intelligence Platform
├── AI gateway
├── Model registry
├── Provider policy
├── Tenant policy
├── Course AI rules
├── Student preferences
├── Data classification engine
├── Consent engine
├── Prompt policy
├── Retrieval pipeline
├── Source authority engine
├── Citation/provenance
├── Agent workflow engine
├── Tool permissions
├── Cost and quota meter
├── Evaluation framework
├── Safety/red-team program
├── Human escalation
├── Override workflow
├── Incident response
├── AI audit log
└── AI transparency interface
```

AI roles to build

Role	Semester AI capability
Student	Copilot, planner, tutor, study assistant, research assistant, writing partner, path explainer
Faculty	Course-authoring assistant, rubric assistant, accessible-content assistant, policy-aware teaching assistant
Advisor	Student-context summarizer, action-plan assistant, support routing, meeting preparation
Registrar	Policy explanation and exception-routing assistant, never final academic authority
Student success	Aggregate trend explanation, workflow suggestions, human-reviewed outreach
Institutional admin	Policy assistant, configuration assistant, evidence/risk summarizer
Security/privacy	Data classification, policy review, AI usage audit
Semester operator	Support summarization, incident assistance, release/evidence search, internal workflow assistant

AI governance checklist

- [ ] Provider registry
- [ ] Model version registry
- [ ] Tenant-approved provider policy
- [ ] Course-level AI policy
- [ ] Academic-integrity boundary
- [ ] Data classification checks
- [ ] Sensitive-data exclusions
- [ ] Consent requirements
- [ ] Source restrictions
- [ ] Approved retrieval sources
- [ ] Citation/provenance requirement
- [ ] Human escalation triggers
- [ ] Spend quota/rate limit
- [ ] Evaluation datasets
- [ ] Bias/safety evaluation
- [ ] Prompt injection defenses
- [ ] Tool permission boundaries
- [ ] AI action audit
- [ ] User correction/feedback loop
- [ ] Incident response procedure
- [ ] Model fallback behavior
- [ ] Explainability UI

- [ ] Retention/deletion policy for AI interactions

7. Campus life and community system

```
Campus Life OS
├── Dining
├── Campus card
├── Housing
├── Room and study-space booking
├── Events
├── Organizations
├── Clubs
├── Volunteer work
├── Student government
├── Recreation
├── Transportation
├── Parking
├── Library
├── Bookstore
├── Campus jobs
├── Local business/off-campus services
├── Community discussions
├── Peer connections
├── Study groups
├── Safety handoffs
├── Wellness resources
└── Accessibility resources
```

Community system requirements

```
Community
├── Community creation
├── Membership
├── Roles
├── Posting
├── Moderation
├── Reports
├── Appeals
├── Alias/pseudonym policy
├── Events
├── Study sessions
├── Media
├── Content retention
├── Safety escalation
├── Volunteer moderation
├── Review calibration
└── Audit
```

The community layer must have:

- Code of conduct.
- Reporting and appeals.

- Human moderation.
- Escalation policy.
- Anti-harassment controls.
- Content retention/deletion rules.
- Strict policy around safety/urgent harm.
- No ungoverned direct messaging for sensitive institutional contexts unless safety and privacy controls are mature.

8. Campus commerce and financial system

Semester may become the student-facing commerce and account layer, but must respect financial controls and payment-provider boundaries.

```
Campus Commerce OS
├── Student account
├── Charges
├── Credits
├── Balances
├── Holds
├── Payment plans
├── Refund status
├── Payments handoff
├── Dining/campus card
├── Bookstore
├── Fees
├── Scholarships
├── Budget planner
├── Financial-aid handoffs
├── Emergency aid handoffs
├── Receipts
├── Reconciliation
├── Dunning/collections workflows
└── Audit ledger
```

Never store raw payment-card data unless you intentionally become a PCI-regulated payment processor—which is unnecessary for Semester. Use approved payment providers and store only required tokens/statuses/reconciliation records.

9. Career, credential, and lifelong learner system

```
Lifelong Education OS
├── Academic record
├── Skills graph
├── Learning evidence
├── Portfolio
├── Credentials
├── Badges
├── Verified achievements
├── Resume
```

```
├── Career planning
├── Opportunities
├── Internships
├── Employer network
├── Alumni mentoring
├── Continuing education
├── Microcredentials
├── Career outcomes
├── Learning pathways
├── Credential wallet
└── Data portability
```

Build toward portable credentials carefully:

```
Learning activity
→ evidence
→ verified skill claim
→ institution/faculty approval where required
→ credential/badge
→ student-controlled sharing
→ employer verification
```

Do not let self-reported claims masquerade as institution-verified credentials.

10. Institutional intelligence and analytics system

Semester must give institutions insight without becoming a student-surveillance platform.

```
Institutional Intelligence
├── Adoption analytics
├── Action completion
├── Service bottlenecks
├── Support trends
├── Registration readiness
├── Course workload signals
├── Capacity planning
├── Demand forecasting
├── Integration health
├── Feature adoption
├── Pilot outcomes
├── Retention proxies
├── Student success outcomes
├── Operational quality
├── Financial/commercial health
├── Security/compliance posture
└── Executive reporting
```

Analytics rules

- Use aggregates by default.
- Enforce minimum cohort size before displaying sensitive group statistics.
- Do not show protected/sensitive data unless policy allows.
- Make metric definitions visible.
- Show data freshness, coverage, and authority.
- Do not use opaque risk scores without explanation.
- Support institutional review/approval for high-impact intervention models.
- Allow audit of how a metric or recommendation was generated.
- Separate correlation from causation.
- Avoid automated punitive decisions.

11. Institutional governance, trust, and compliance system

```
Education Governance OS
├── Tenant hierarchy
├── Policy hierarchy
├── Stewardship assignments
├── Configuration studio
├── Workflow builder
├── Feature flags
├── Rollouts
├── Approvals
├── Two-person controls
├── Break-glass access
├── Audit events
├── Data classification
├── Consent
├── Retention
├── Legal holds
├── Data requests
├── Privacy
├── Security controls
├── Compliance framework mapping
├── Trust Center
├── HECVAT evidence
├── Accessibility evidence
├── Claims governance
├── Vendor governance
├── Incident management
├── Business continuity
├── Disaster recovery
└── Offboarding and portability
```

Trust Center system

```
Trust Center
├── Security overview
├── Privacy overview
├── Accessibility statement
├── Responsible AI
├── Data practices
├── Subprocessors
├── Availability and status
├── Incident communication
├── Compliance roadmap
├── HECVAT materials
├── DPA process
├── Security questionnaire request
├── Trust room
├── Vulnerability reporting
└── Customer document access
```

12. Platform ecosystem and marketplace

Semester becomes a platform when others can safely build on it.

```
Semester Developer Platform
├── API gateway
├── OAuth/OIDC
├── API keys
├── Service accounts
├── Webhooks
├── Event subscriptions
├── SDKs
├── Developer docs
├── Sandbox tenants
├── Test data
├── Integration certification
├── Marketplace
├── App review
├── Partner policies
├── Revenue share
├── Billing metering
├── Extension permissions
├── Tenant install controls
├── Versioning
├── Deprecation policy
└── Incident notifications
```

Marketplace categories

- Learning tools
- Accessibility tools
- Tutoring

- Career services
- Library/research services
- Campus services
- Advising services
- Financial wellness
- Scheduling
- Student organizations
- Institutional analytics
- AI models/providers
- Publisher/courseware
- Identity/integration tools
- Developer extensions

Every marketplace app must have:

```
Publisher identity
Security review
Privacy review
Data scopes
Tenant install approval
Permission request
Version history
Support owner
Pricing model
Incident contact
Offboarding/export process
```

13. Semester company operating system

Semester must run itself with the same quality it sells.

```
Semester Company OS
├── Corporate governance
├── Finance
├── Budget/runway
├── Pricing
├── Billing
├── Sales
├── GTM
├── Marketing
├── Customer success
├── Implementation
├── Support
├── Product operations
├── Engineering
├── Security
├── Compliance
```

```
├── Legal operations
├── Vendor management
├── Partnerships
├── People and hiring
├── Board/investor reporting
├── Risk management
├── Incident operations
├── Release operations
├── Data governance
├── AI governance
└── Operations Command Center
```

Internal company dashboards

```
Executive dashboard
├── Revenue
├── Pipeline
├── Pilots
├── Conversion
├── Product adoption
├── Customer health
├── Churn risk
├── Support
├── Reliability
├── Security
├── Compliance
├── Cash runway
├── Hiring capacity
└── Risks requiring decision
```

14. Required operating functions

Function	Must own
Product	Roadmap, requirements, outcomes, research, feature decisions
Engineering	Architecture, quality, performance, reliability, security implementation
Design	User research, interaction, accessibility, design system
Learning	Pedagogy, course design, learning evidence, assessment integrity
Academic operations	Registrar, records, policies, degree progression, institutional rules
Student success	Advising workflows, support, interventions, outcomes
Campus services	Service integrations, student experience, operational handoffs
AI governance	Model policy, evaluation, provider controls, safety, spend
Security	Access, vulnerabilities, incident response, evidence
Privacy/legal	Data policy, contracts, retention, requests, legal holds
GTM	ICP, messaging, funnel, pilots, pricing, pipeline

Function	Must own
Customer success	Onboarding, implementation, health, renewal, expansion
Finance	Pricing, budget, collections, runway, reporting
Partnerships	Integrations, marketplace, channels, ecosystem
People	Hiring, culture, access lifecycle, training
Operations	Cadence, risks, decisions, execution, reporting

15. Full operating cadence

Cadence	Review	Core questions
Daily	Operations triage	What is broken, blocked, urgent, or customer-impacting?
Weekly	Product/engineering	What shipped, failed, learned, and needs decision?
Weekly	Revenue/pilot	What accounts moved, what is blocked, what closes next?
Weekly	Customer success	Which pilots/customers are at risk or ready to expand?
Weekly	Security/reliability	What changed, what is vulnerable, what is consuming error budget?
Monthly	Finance	What is runway, spend, margin, collections, and forecast variance?
Monthly	Trust/compliance	What evidence, policy, vendor, or review is due?
Monthly	Leadership	What are the highest-leverage decisions and risks?
Quarterly	Strategy/OKR	What to stop, continue, fund, hire, replace, or scale?
Quarterly	Board/investor	What is true about traction, risks, capital, and strategic choices?

16. Full release and replacement gate

No feature, domain, migration, or institutional capability is complete until it has:

```
1. Product requirement
2. Design specification
3. Accessibility review
4. Data classification
5. Permission model
6. Tenant isolation
7. Source authority
8. Audit model
9. Security threat model
10. Privacy/retention plan
11. AI policy, if AI is involved
12. Integration plan
13. Migration plan
14. Rollback plan
15. Support playbook
16. Monitoring/SLO
17. Test suite
18. Documentation
```

```
19. Release evidence
20. Owner and review date
```

Ultimate Semester command

Use this as the broadest possible execution prompt. It expands the previous architecture and instructs an agent to build, audit, prioritize, and operationalize the full company/platform.

```
You are the complete executive, technical, product, learning, operations,
security, governance, commercial, and platform leadership team for Semester.

Repository:
https://github.com/harrisonjrubin7-cmyk/semester

Semester is not a point solution.

Semester is the unified, AI-native operating system for education.

Its mission is to replace the fragmented infrastructure of:

- learning
- courses
- LMS
- SIS/ERP
- registrar
- academic records
- registration
- degree audit
- advising
- student success
- support
- productivity
- calendar
- tasks
- notes
- files
- collaboration
- search
- AI assistants
- research
- assessment
- gradebook
- student accounts
- payment plans
- financial workflows
- campus services
- housing
- dining
- events
- organizations
- community
- safety handoffs
- accessibility services
- career systems
```

```
- alumni
- employer networks
- family/guardian systems
- marketplace partners
- developer ecosystem
- institutional governance
- trust/compliance
- company operations

Semester must become:

- Student OS
- Productivity suite
- Learning OS
- Course Studio
- AI intelligence platform
- Registrar and academic OS
- Student success OS
- Campus super-app
- Financial/account OS
- Career and lifelong-learning OS
- Family/guardian consent platform
- Institutional operating system
- Developer platform
- Marketplace
- Company operating system

The core doctrine is:

BUILD EVERYTHING NATIVELY.
INTEGRATE RESPONSIBLY.
REPLACE ONLY WITH PROOF.
OPERATE EVERY DOMAIN AS ONE SYSTEM.

Do not narrow the thesis.

However, do not make false claims that a native replacement exists before it is
implemented, tested, accessible, secure, governed, migrated, approved, and
supportable.

Do not replace official institutional records, identity, registration, grades,
financial accounts, legal processes, or high-stakes decisions without formal
institutional authority, migration, reconciliation, audit, rollback, and
operational readiness.

============================================================
YOUR FIRST TASK: CREATE THE MASTER OPERATING MAP
============================================================

Audit the complete current Semester repository, database, migrations,
documentation, code, feature inventory, marketing, security, operations, GTM,
customer success, integrations, and deployment system.

Create:

docs/master/SEMESTER_COMPLETE_OPERATING_SYSTEM.md
```

```
docs/master/SEMESTER_PRODUCT_UNIVERSE.md
docs/master/SEMESTER_DOMAIN_CATALOG.md
docs/master/SEMESTER_ROLE_CATALOG.md
docs/master/SEMESTER_CAPABILITY_MATRIX.md
docs/master/SEMESTER_DATA_AUTHORITY_MATRIX.md
docs/master/SEMESTER_INTEROPERABILITY_MATRIX.md
docs/master/SEMESTER_DOMAIN_REPLACEMENT_GATES.md
docs/master/SEMESTER_MIGRATION_FACTORY.md
docs/master/SEMESTER_TRUST_AND_GOVERNANCE_MODEL.md
docs/master/SEMESTER_COMPANY_OPERATING_SYSTEM.md
docs/master/SEMESTER_PLATFORM_ECOSYSTEM.md
docs/master/SEMESTER_OUTCOME_MEASUREMENT.md
docs/master/SEMESTER_RISK_REGISTER.md
docs/master/SEMESTER_12_MONTH_EXECUTION_PLAN.md
docs/master/SEMESTER_MASTER_BACKLOG.md

For every current and planned capability, classify:

- Native and verified
- Native but incomplete
- Integrated
- Transitional
- Pilot-only
- Designed/documented
- Planned
- Not started
- Requires security review
- Requires privacy/legal review
- Requires accessibility review
- Requires institutional approval
- Requires migration
- Requires reconciliation
- Requires rollback
- Unsafe to activate
- Ready for pilot
- Ready for production
- Ready to become authoritative system of record

============================================================
PRODUCT DOMAINS TO MAP, DESIGN, AND BUILD
============================================================

1. Student OS
2. Workspace/productivity
3. Path and degree planning
4. Course Studio/LMS
5. Learning evidence and assessment
6. AI gateway and copilot
7. Search and knowledge graph
8. Faculty experience
9. Advisor/student success
10. Registrar and academic operations
11. Academic records and grade ledger
12. Registration and enrollment
13. Student finance/account/payment plans
14. Financial aid and scholarship handoffs
```

15. Campus life/services
16. Housing
17. Dining
18. Events
19. Community and organizations
20. Accessibility services
21. Safety and emergency handoffs
22. Library/research
23. Career/employer/alumni
24. Family/guardian grants
25. Institutional governance
26. Identity/SSO/SCIM
27. Integrations/LTI/OneRoster/Edu-API
28. Privacy/retention/legal holds
29. Security/audit/incident response
30. Trust/compliance/HECVAT
31. Operations Command Center
32. Commercial/billing/customer success
33. Marketing/sales/company site
34. Developer platform
35. Marketplace/partners
36. Data/analytics/outcomes
37. Reliability/SLO/release operations
38. People/hiring/company operations
39. Finance/runway/board reporting
40. Globalization/localization/accessibility expansion

For every domain define:

- Vision
- Primary users
- User jobs to be done
- Current source of truth
- Native Semester source of truth
- Integration strategy
- Data model
- Tenant scope
- Role/capability model
- Data classification
- Consent model
- AI model/policy
- Accessibility requirements
- Security requirements
- Audit requirements
- Privacy/retention requirements
- Migration plan
- Reconciliation plan
- Dual-run plan
- Cutover criteria
- Rollback plan
- Support model
- Incident model
- SLO/reliability requirements
- Commercial model
- Customer success model
- Outcome metrics

- Required documentation
- Release gates
- Replacement authority gates
- Team owner
- Dependencies
- Risks
- Priority
- Implementation branch
- Tests

==
IMPLEMENTATION PROGRAM
==

PHASE 0:
Establish current truth. No unsupported claims. No production-impacting changes.

PHASE 1:
Identity, tenancy, capability, RLS, audit, data classification, consent, retention, integrations, rollout controls, observability, support baseline.

PHASE 2:
Student OS:
Home, Today, Action Center, Path, Plan, Calendar, Tasks, Notes, Search, Workspace, Copilot, Support, Career.

PHASE 3:
Learning OS:
Course Studio, resources, study packs, objectives, assessments, feedback, gradebook, integrity, course AI policy, learning evidence.

PHASE 4:
Student Success:
Advising, check-ins, referrals, success plans, support, institutional actions, outcome analytics.

PHASE 5:
Academic Core:
Catalog, terms, sections, enrollment, registration, holds, prerequisites, overrides, records, grade ledger, degree planning, transfer, graduation.

PHASE 6:
Campus Services:
Dining, housing, events, community, organizations, spaces, accessibility, campus employment, support, safe handoffs.

PHASE 7:
Career/Lifelong:
Skills graph, portfolio, credential wallet, employer network, mentoring, opportunities, alumni and continuing education.

PHASE 8:
Institutional OS:
Configuration, policy, workflow builder, integrations, trust, compliance, privacy, security, rollout, offboarding.

```
PHASE 9:
Company OS:
GTM, pilots, billing, customer success, support, renewals, finance,
operations, people, partnerships, board reporting.

PHASE 10:
Platform Ecosystem:
APIs, webhooks, OAuth/OIDC, developer portal, SDKs, sandbox tenants,
marketplace, partner certification, extension framework.

============================================================
NON-NEGOTIABLE CONTROLS
============================================================

Every feature/domain requires:

- Source authority
- Tenant isolation
- Role/capability check
- Data classification
- Consent where appropriate
- Accessibility requirement
- Security threat model
- Privacy/retention plan
- Audit event
- Support route
- Monitoring/SLO
- Test coverage
- Migration/rollback plan where relevant
- Documentation
- Owner
- Review date
- Release evidence

AI must be:

- Policy-bound
- Course-aware
- Tenant-aware
- Permission-aware
- Source-aware
- Citation/provenance-aware
- Cost-aware
- Integrity-aware
- Human-escalation-capable
- Audited
- Evaluated
- Never silently authoritative for official institutional decisions

============================================================
FINAL REQUIRED OUTPUT
============================================================

Return:

1. Complete architecture map
```

```
2. Complete domain catalog
3. Capability matrix
4. Role matrix
5. Data authority matrix
6. Integration and migration factory
7. Domain replacement gates
8. Security/privacy/accessibility/control model
9. Company operating system
10. Platform ecosystem plan
11. Outcomes and moat plan
12. 12-month roadmap
13. P0/P1/P2/P3 backlog
14. Required team/ownership model
15. Required external systems
16. Test strategy
17. Release strategy
18. Budget/capacity model
19. Risk register
20. Exact next 25 actions in order

Do not stop at a vision document. Produce implementation-ready artifacts,
schemas, APIs, screens, tests, migrations, workflows, operating procedures,
and traceability records.
```

Final test

Every time you consider adding a module, ask:

```
Does this make Semester more like one coherent education operating system,
or does it create another disconnected app inside Semester?
```

If it creates a disconnected app, it must be redesigned until it connects to the shared identity, path, action, policy, evidence, trust, and operations layers.

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1. expand-evan-further-provide-prompt-and-comand-to-e.pdf